Assignment 4 Packet Tracer - TCP and UDP Communications
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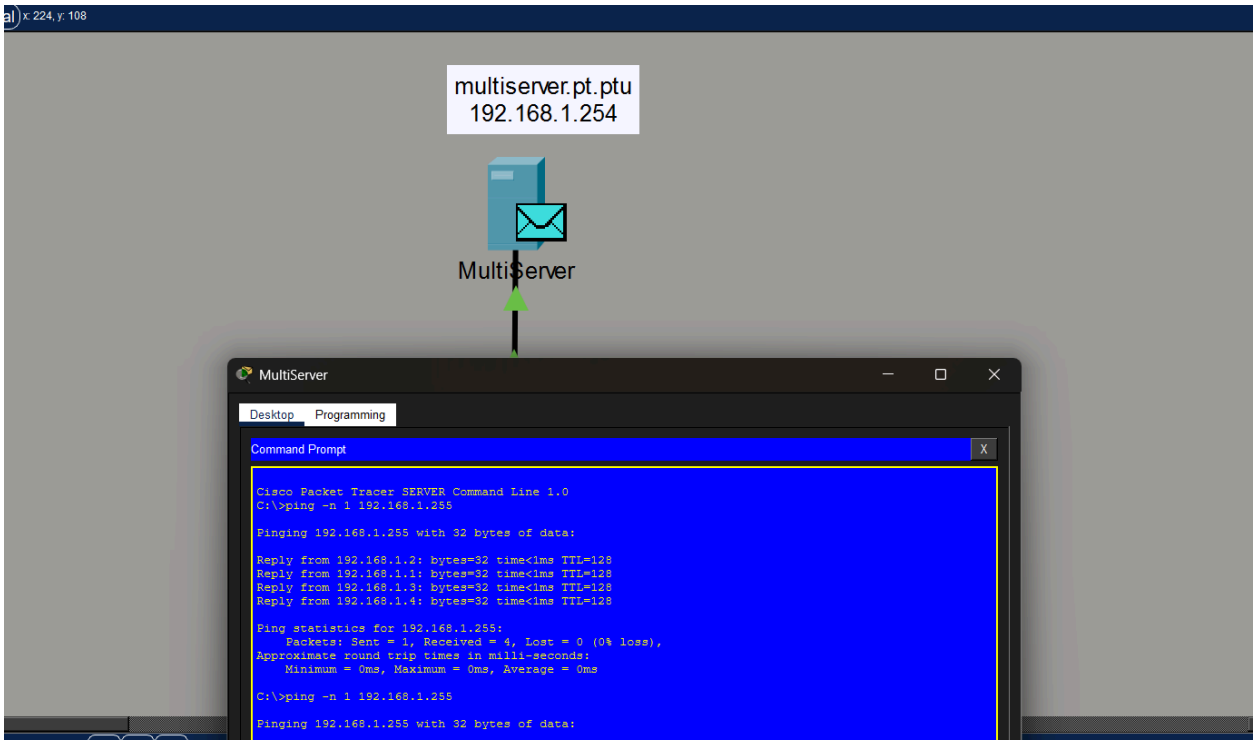
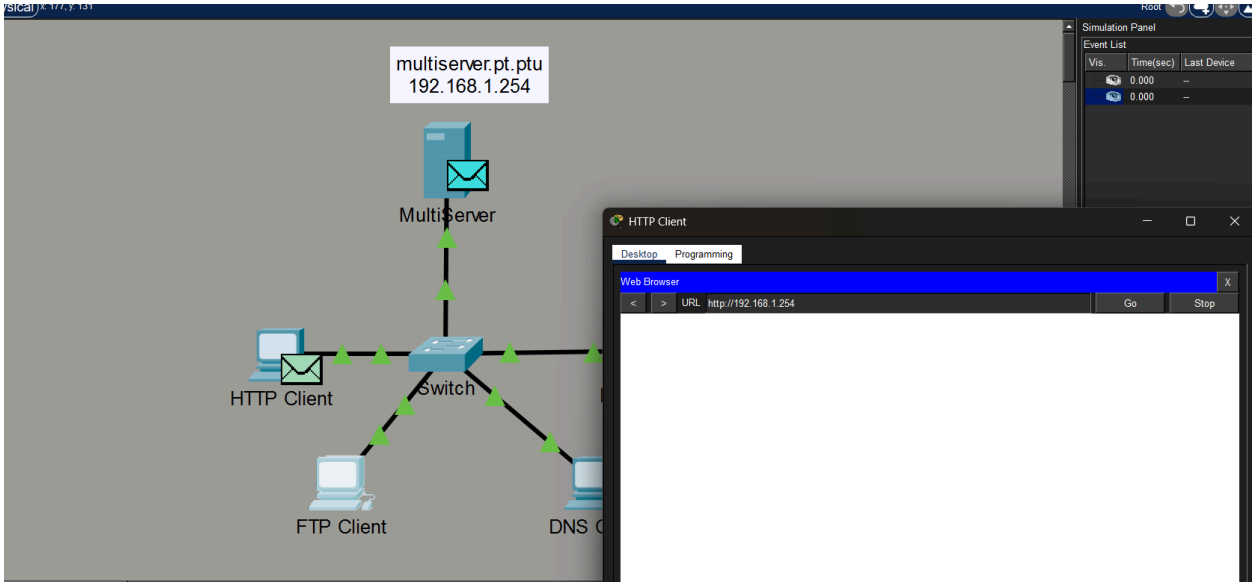
Part 1: Generate Network Traffic in Simulation Mode and View Multiplexing
Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.
a. Click MultiServer and click the Desktop tab > Command Prompt.
b. Enter the ping -n 1 192.168.1.255 command. You are pinging the broadcast address for the client LAN.
The command option will send only one ping request rather than the usual four. This will take a few
seconds as every device on the network responds to the ping request from MultiServer.
c. Close the MultiServer window.



Step 2: Generate web (HTTP) traffic.
a. Switch to Simulation mode.
b. Click HTTP Client and open the Web Browser from the desktop.
c. In the URL field, enter 192.168.1.254 and click Go. Envelopes (PDUs) will appear in the topology window.
d. Minimize, but do not close, the HTTP Client configuration window.

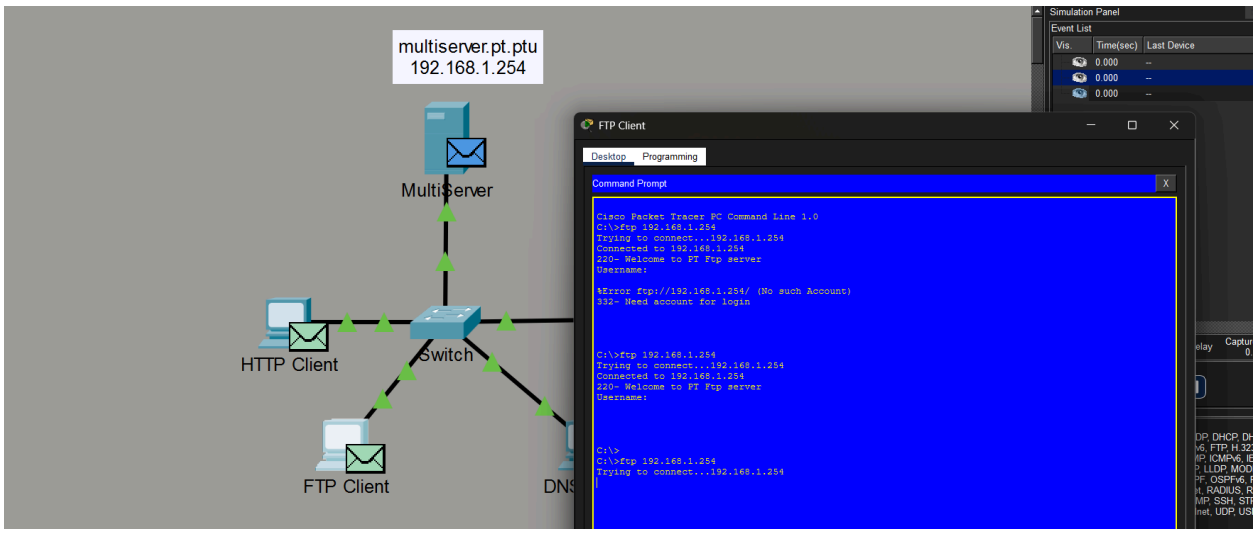


Step 3: Generate FTP traffic.

a. Click FTP Client and open the Command Prompt from the desktop

b. Enter the ftp 192.168.1.254 command. PDUs will appear in the simulation window.

c. Minimize, but do not close, the FTP Client configuration window.

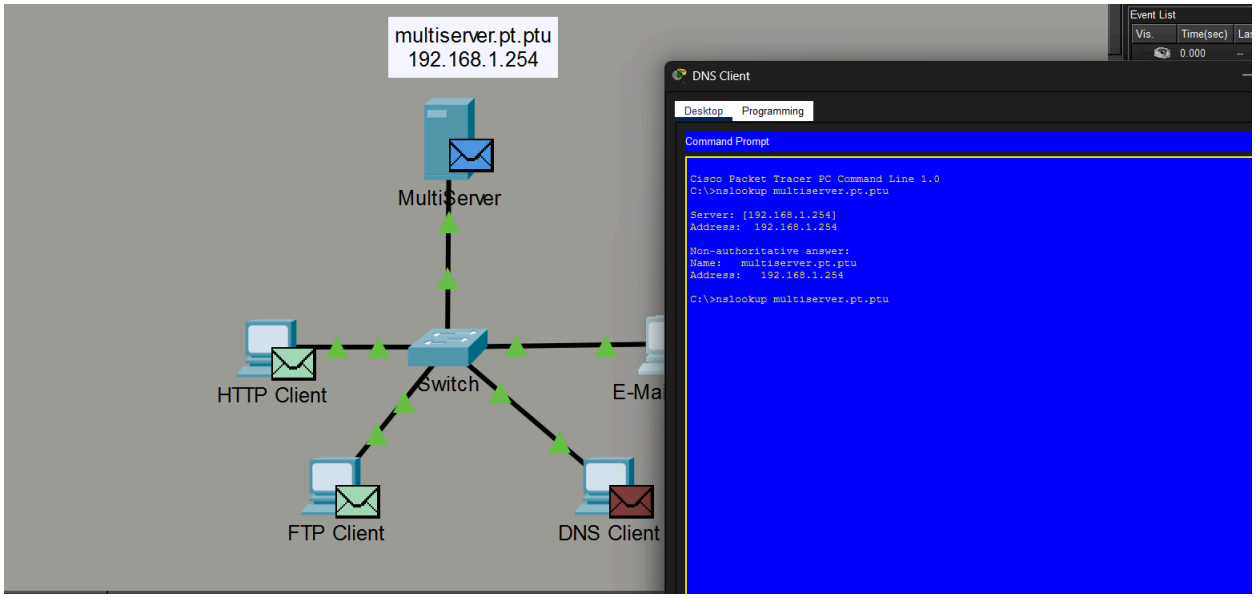


Step 4: Generate DNS traffic.

a. Click DNS Client and open the Command Prompt.

b. Enter the nslookup multiserver.pt.ptu command. A PDU will appear in the simulation window.

c. Minimize, but do not close, the DNS Client configuration window.



Step 5: Generate Email traffic.

a. Click E-Mail Client and open the E Mail tool from the Desktop.

b. Click Compose and enter the following information:

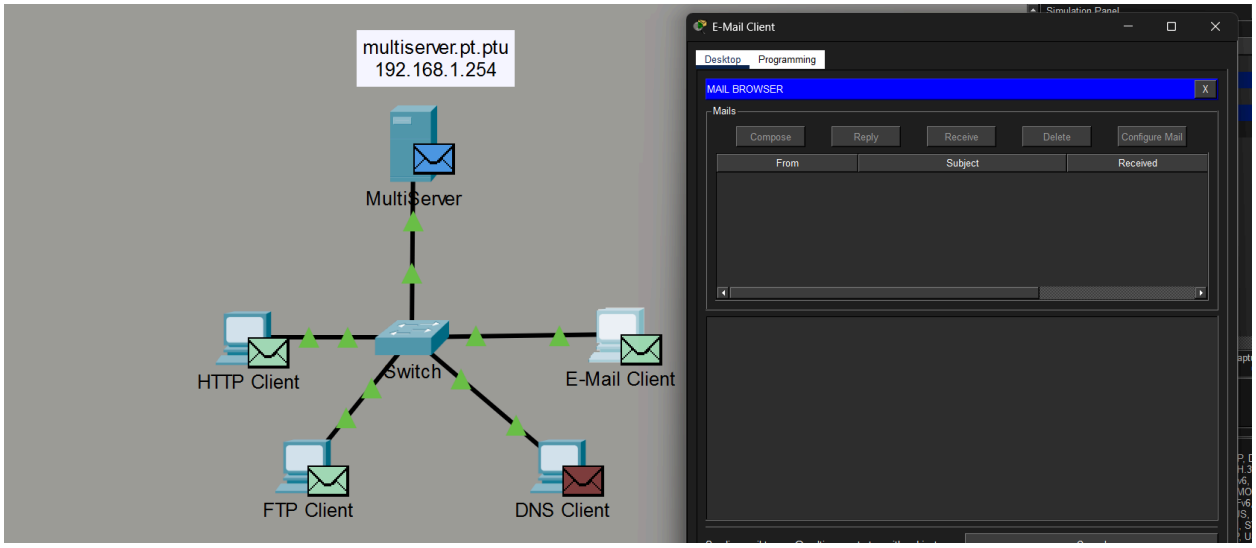
1) To: user@multiserver.pt.ptu

2) Subject: personalize the subject line

3) E-Mail Body: personalize the Email

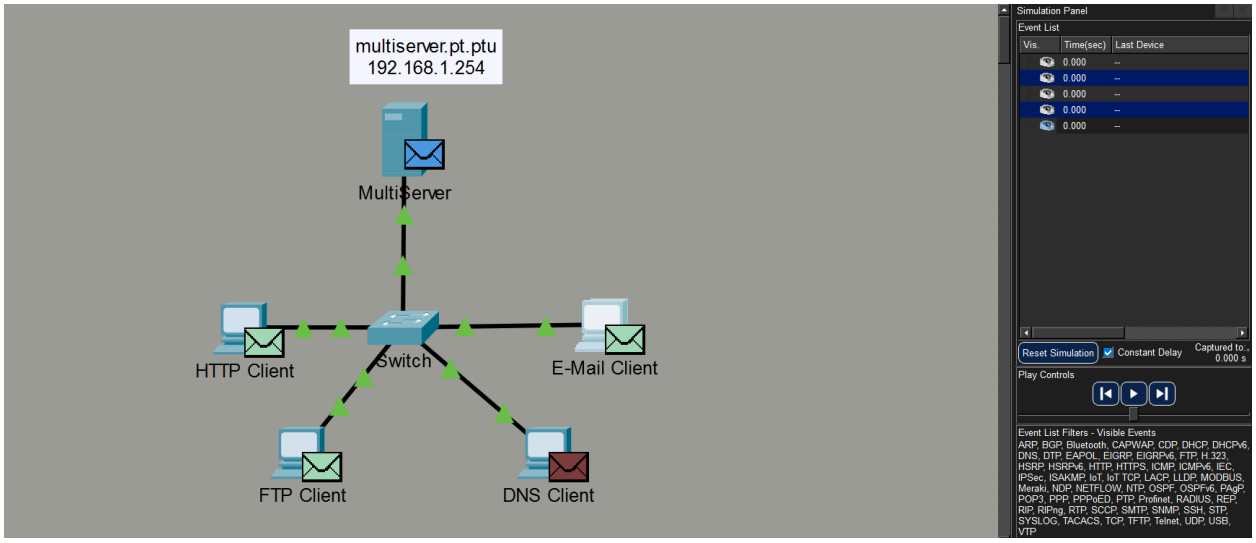
c. Click Send.

d. Minimize, but do not close, the E-Mail Client configuration window.



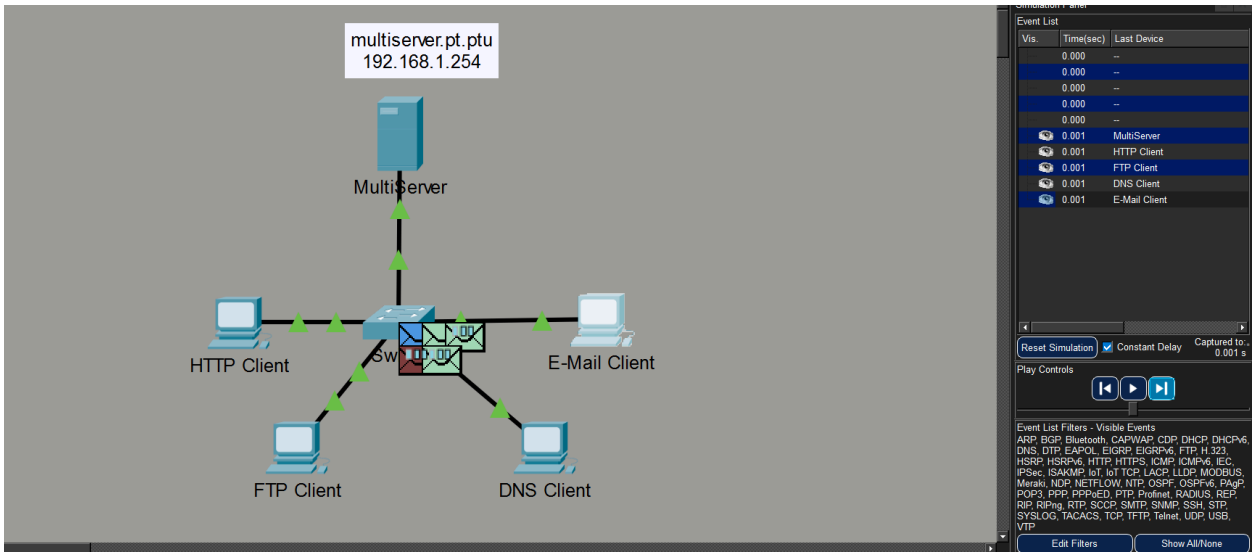
Step 6: Verify that the traffic is generated and ready for simulation.

There should now be PDU entries in the simulation panel for each of the client computers.

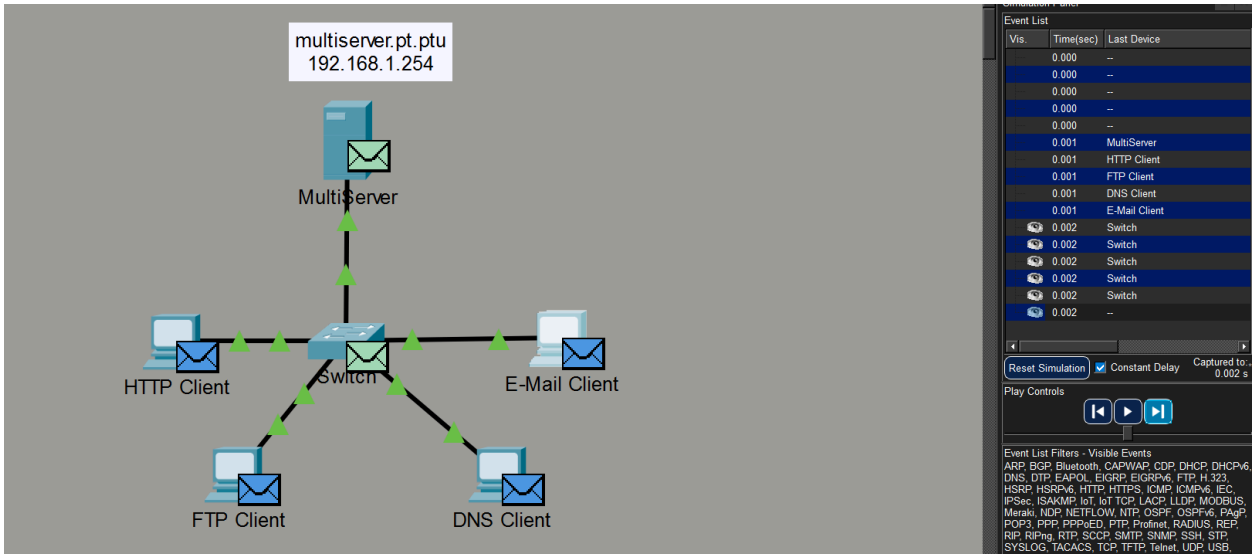


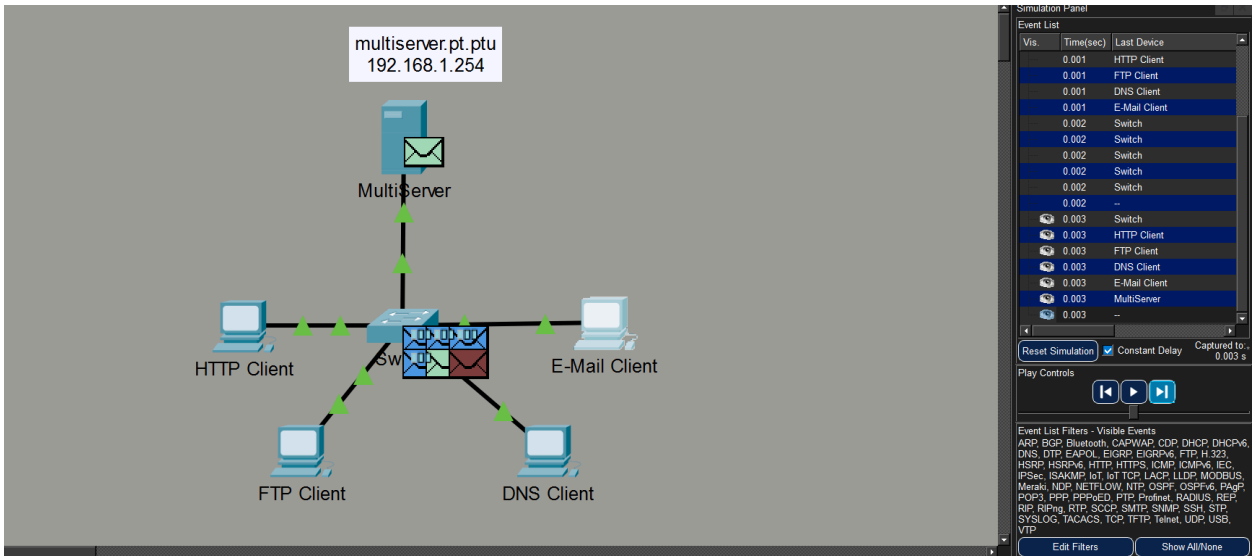
Step 7

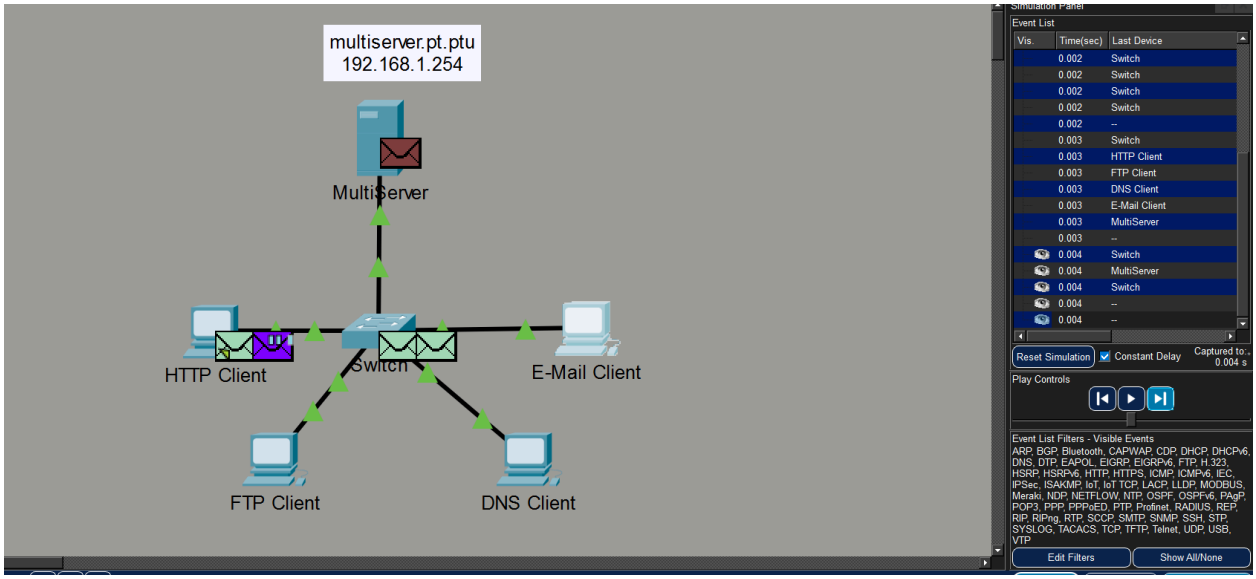
a. Click Capture/Forward once. All of the PDUs travel to the switch.

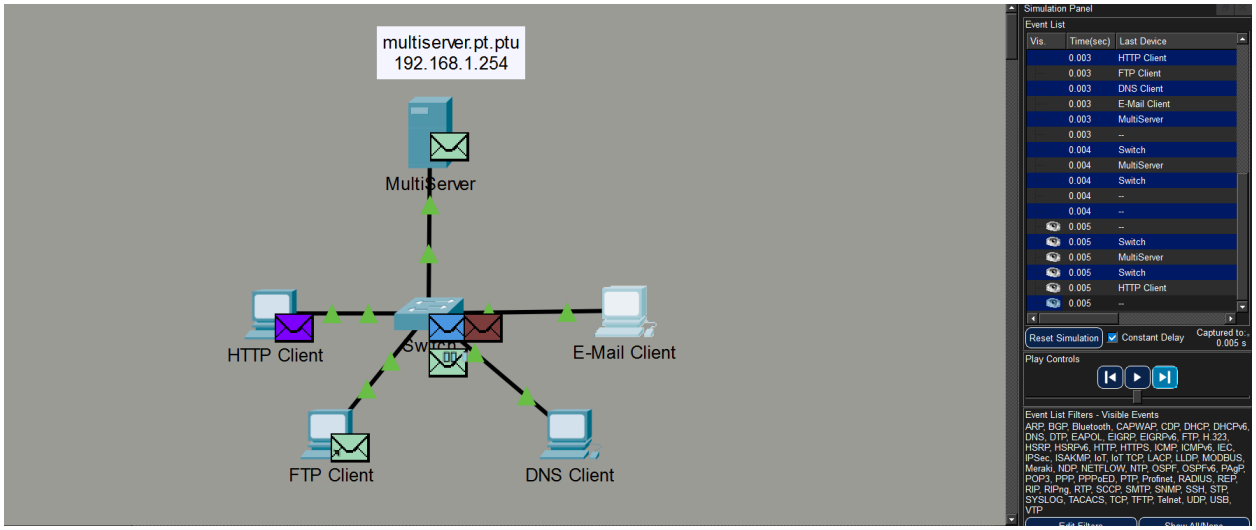


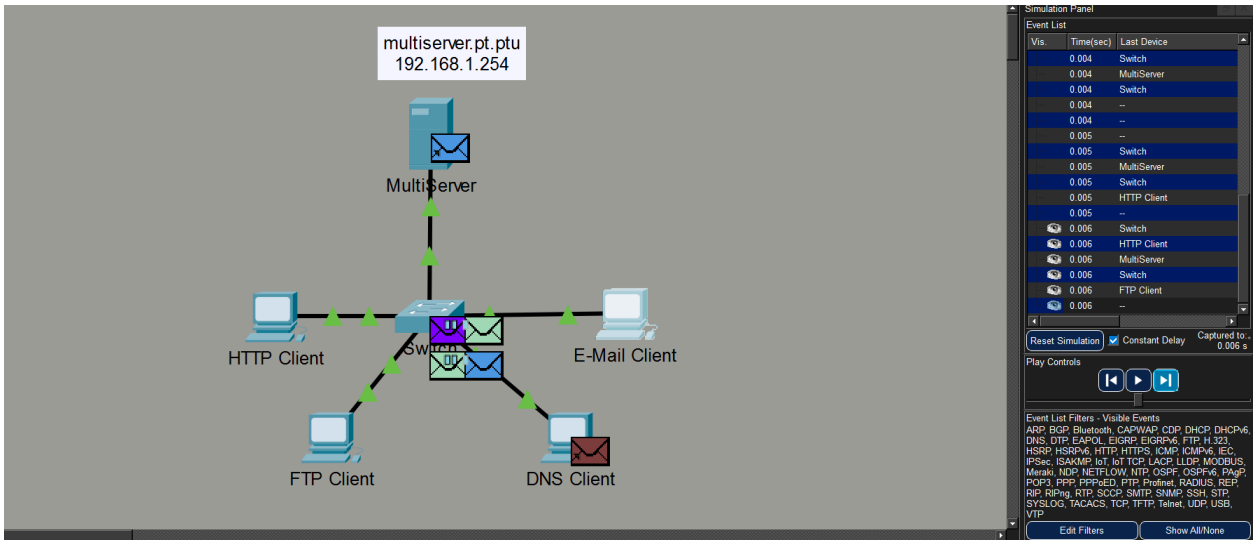
b. Click Capture/Forward six times and watch the PDUs from the different hosts as they travel on the network.

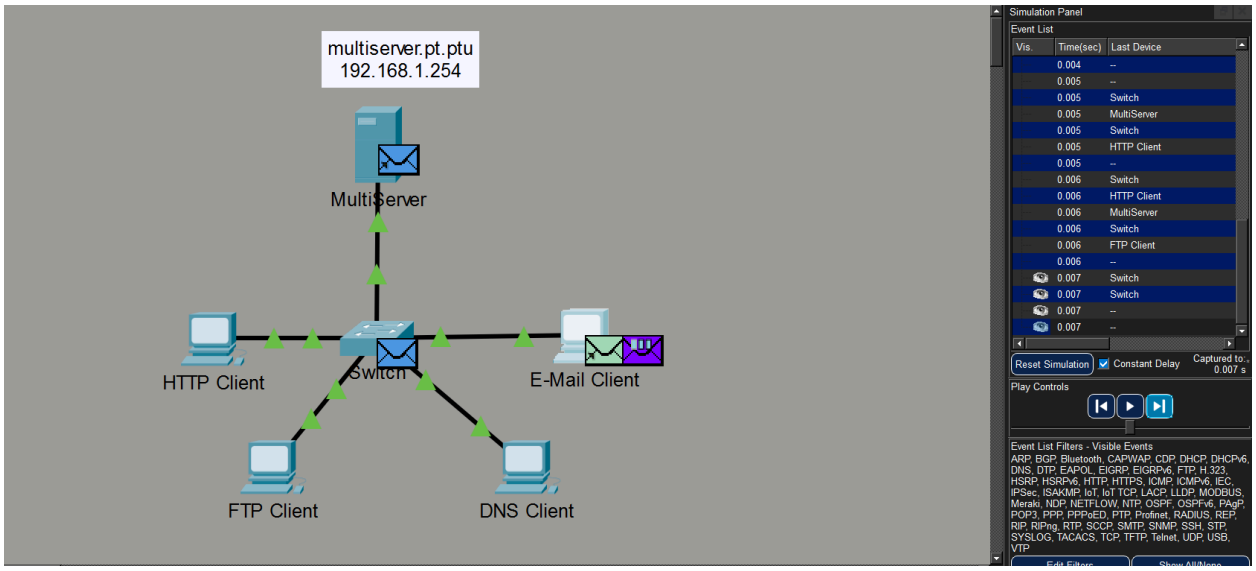

multiserver.pt.ptu
192.168.1.254
MultiServer
HTTP Client
Switch
E-Mail Client
FTP Client
DNS Client
Event List
Vis. Time(sec) Last Device
0.000 --
0.000 --
0.000 --
0.000 --
0.000 --
0.001 MultiServer
0.001 HTTP Client
0.001 FTP Client
0.001 DNS Client
0.001 E-Mail Client
0.002 Switch
0.002 Switch
0.002 Switch
0.002 Switch
0.002 Switch
0.002 --
Reset Simulation
Constant Delay
Captured to: 0.002 s
Play Controls
Event List Filters - Visible Events
ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB,


multiserver.pt.ptu
192.168.1.254
MultiServer
HTTP Client
E-Mail Client
FTP Client
DNS Client
Simulation Panel
Event List
Vis. Time(sec) Last Device
0.001 HTTP Client
0.001 FTP Client
0.001 DNS Client
0.001 E-Mail Client
0.002 Switch
0.002 Switch
0.002 Switch
0.002 Switch
0.002 Switch
0.002 --
0.003 Switch
0.003 HTTP Client
0.003 FTP Client
0.003 DNS Client
0.003 E-Mail Client
0.003 MultiServer
0.003 --
Reset Simulation
Constant Delay
Captured to: 0.003 s
Play Controls
Event List Filters - Visible Events
ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None


multiserver.pt.ptu
192.168.1.254
MultiServer
HTTP Client
Switch
E-Mail Client
FTP Client
DNS Client
Simulation Panel
Event List
Vis. Time(sec) Last Device
0.002 Switch
0.002 Switch
0.002 Switch
0.002 Switch
0.002 --
0.003 Switch
0.003 HTTP Client
0.003 FTP Client
0.003 DNS Client
0.003 E-Mail Client
0.003 MultiServer
0.003 --
0.004 Switch
0.004 MultiServer
0.004 Switch
0.004 --
0.004 --
Reset Simulation
Constant Delay
Captured to: 0.004 s
Play Controls
Event List Filters - Visible Events
ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None


multiserver.pt.ptu
192.168.1.254
MultiServer
HTTP Client
Switch
E-Mail Client
FTP Client
DNS Client
Simulation Panel
Event List
Vis. Time(sec) Last Device
0.003 HTTP Client
0.003 FTP Client
0.003 DNS Client
0.003 E-Mail Client
0.003 MultiServer
0.003 --
0.004 Switch
0.004 MultiServer
0.004 Switch
0.004 --
0.004 --
0.005 --
0.005 Switch
0.005 MultiServer
0.005 Switch
0.005 HTTP Client
0.005 --
Reset Simulation
Constant Delay
Captured to: 0.005 s
Play Controls
Event List Filters - Visible Events
ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP


multiserver.pt.ptu
192.168.1.254
MultiServer
HTTP Client
Switch
E-Mail Client
FTP Client
DNS Client
Simulation Panel
Event List
Vis. Time(sec) Last Device
0.004 Switch
0.004 MultiServer
0.004 Switch
0.004 --
0.004 --
0.005 --
0.005 Switch
0.005 MultiServer
0.005 Switch
0.005 HTTP Client
0.005 --
0.006 Switch
0.006 HTTP Client
0.006 MultiServer
0.006 Switch
0.006 FTP Client
0.006 --
Reset Simulation
Constant Delay
Captured to: 0.006 s
Play Controls
Event List Filters - Visible Events
ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None

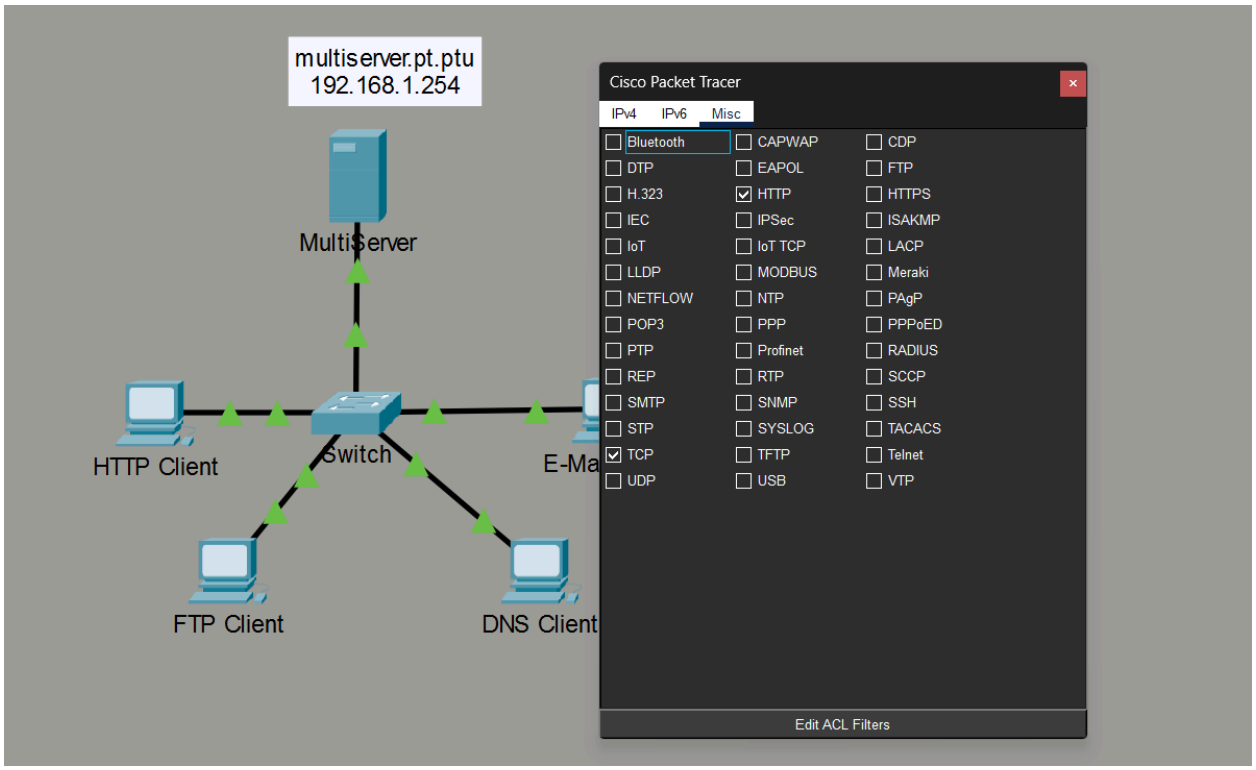

multiserver.pt.ptu
192.168.1.254
MultiServer
HTTP Client
Switch
E-Mail Client
FTP Client
DNS Client
Simulation Panel
Event List
Vis. Time(sec) Last Device
0.004 --
0.005 --
0.005 Switch
0.005 MultiServer
0.005 Switch
0.005 HTTP Client
0.005 --
0.006 Switch
0.006 HTTP Client
0.006 MultiServer
0.006 Switch
0.006 FTP Client
0.006 --
0.007 Switch
0.007 Switch
0.007 --
0.007 --
Reset Simulation
Constant Delay
Captured to: 0.007 s
Play Controls
Event List Filters - Visible Events
ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

What is this called?
Multiplexing หลาย ๆ การสื่อสารถูกส่งผ่านสายเดียวกัน โดยแยกกันด้วย port number

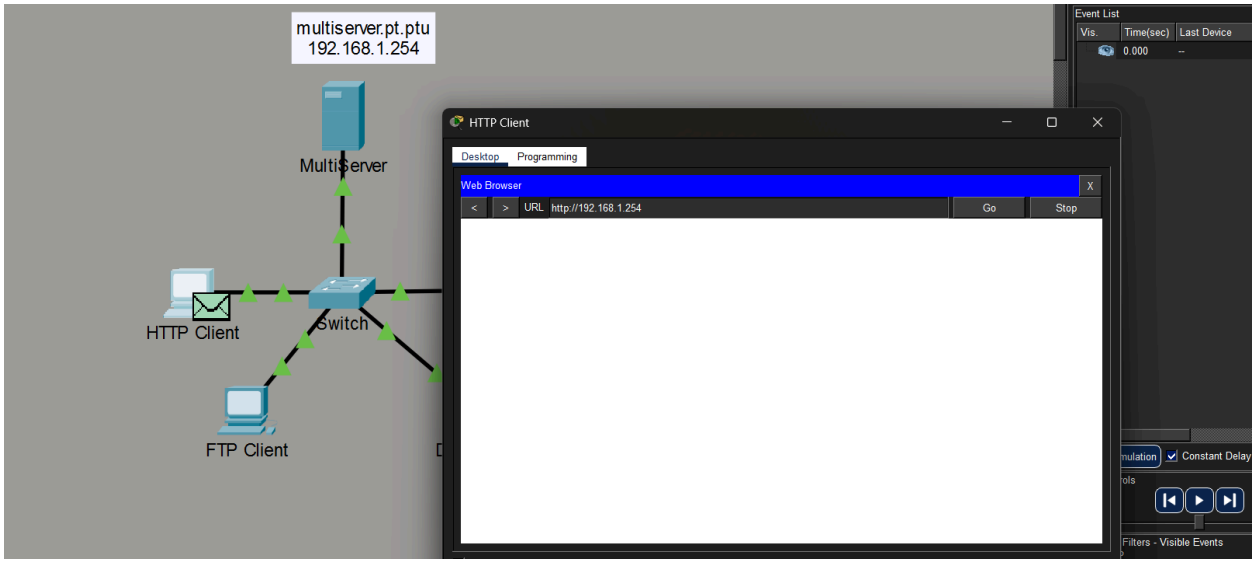
A variety of PDUs appears in the event list in the Simulation Panel. What is the meaning of the different colors?
สีแทน protocol ที่แตกต่างกัน เช่น HTTP, FTP, DNS, TCP, UDP เพื่อให้แยกชนิดของ PDU ได้ง่าย

Part 2: Examine Functionality of the TCP and UDP Protocols
Step 1: Examine HTTP traffic as the clients communicate with the server.
a. Click Reset Simulation.
b. Filter the traffic that is currently displayed to only HTTP and TCP PDUs. To filter the traffic that is currently displayed:

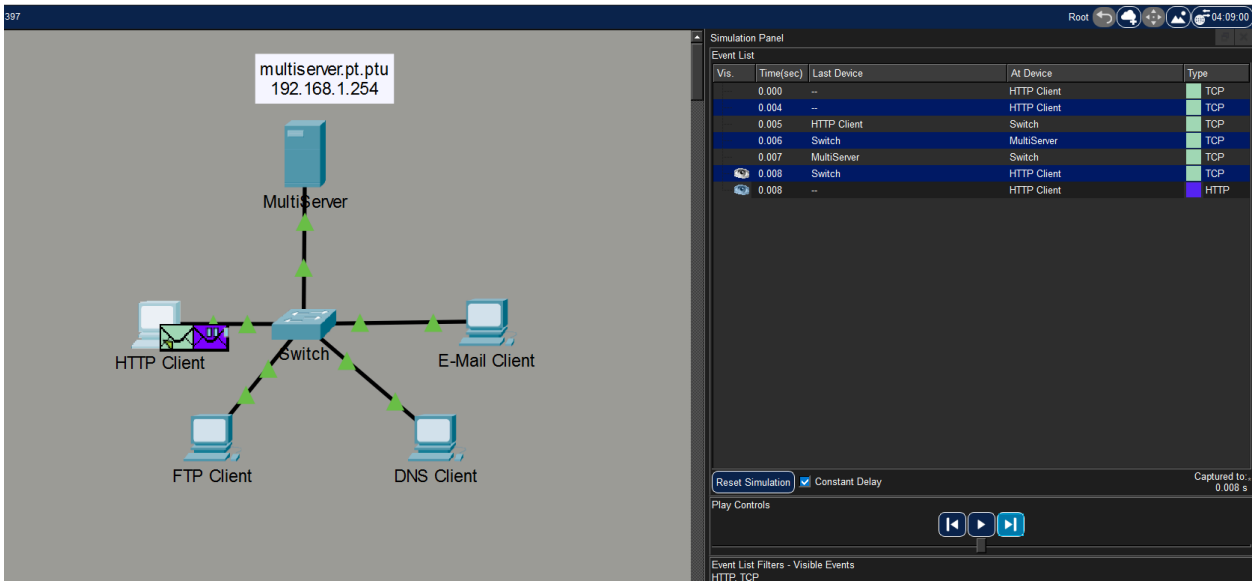
1) Click Edit Filters and toggle the Show All/None button.
2) Select HTTP and TCP. Click the red “x” in the upper right-hand corner of the Edit Filters box to close it. Visible Events should now display only HTTP and TCP PDUs.



c. Open the browser on HTTP Client and enter 192.168.1.254 in the URL field. Click Go to connect to the server over HTTP. Minimize the HTTP Client window.

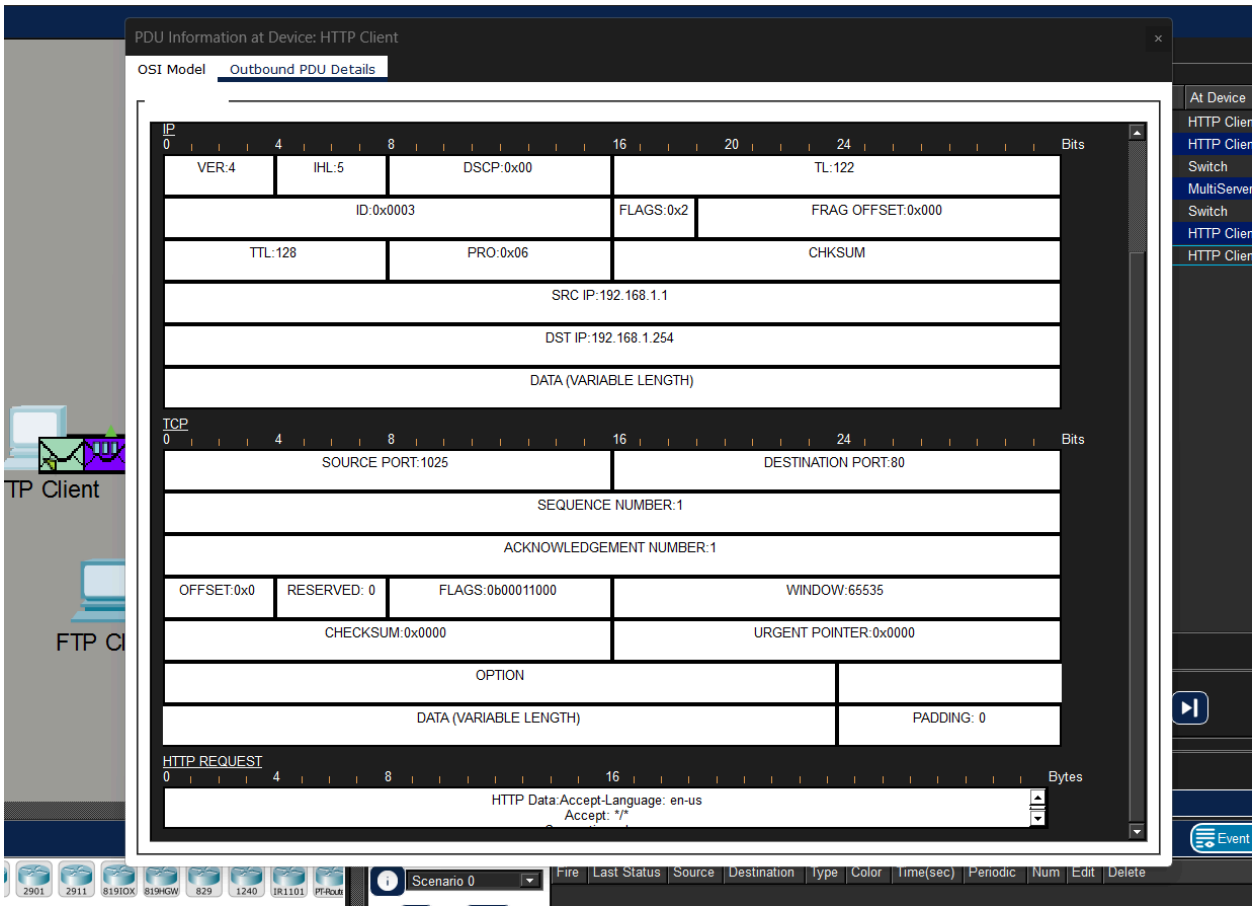


d. Click Capture/Forward until you see a PDU appear for HTTP. Note that the color of the envelope in the topology window matches the color code for the HTTP PDU in the Simulation Panel.



Question:
Why did it take so long for the HTTP PDU to appear?
เพราะต้องมี **TCP 3-way handshake (SYN → SYN-ACK → ACK)** ก่อนถึงจะเริ่มส่งข้อมูล HTTP

e. Click the PDU envelope to show the PDU details. Click the Outbound PDU Details tab and scroll down to the second to the last section.



What is the section labeled?
TCPI
Are these communications considered to be reliable?
ใช่ เพราะ มี sequence number กับ acknowledgement number

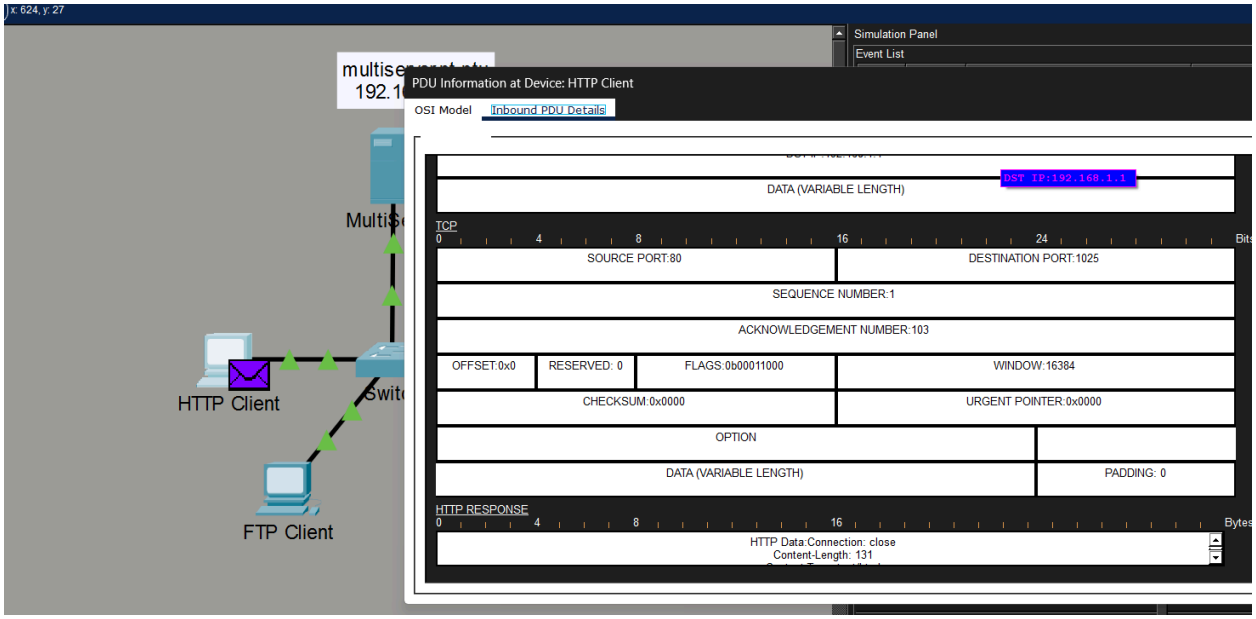
Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.
SRC PORT : 1025
DEST PORT : 80
SEQUENCE NUM : 1
ACK NUM : 1

f. Look at the value in the Flags field, which is located next to the Window field. The values to the right of
the “b” represent the TCP flags that are set for this stage of the data conversation. Each of the six places corresponds to a flag. The presence of a “1” in any place indicates that the flag is set. More than one flag can be set at a time. The values for the flags are shown below.

Flag Place	6	5	4	3	2	1
Value	URG	ACK	PSH	RST	SYN	FIN

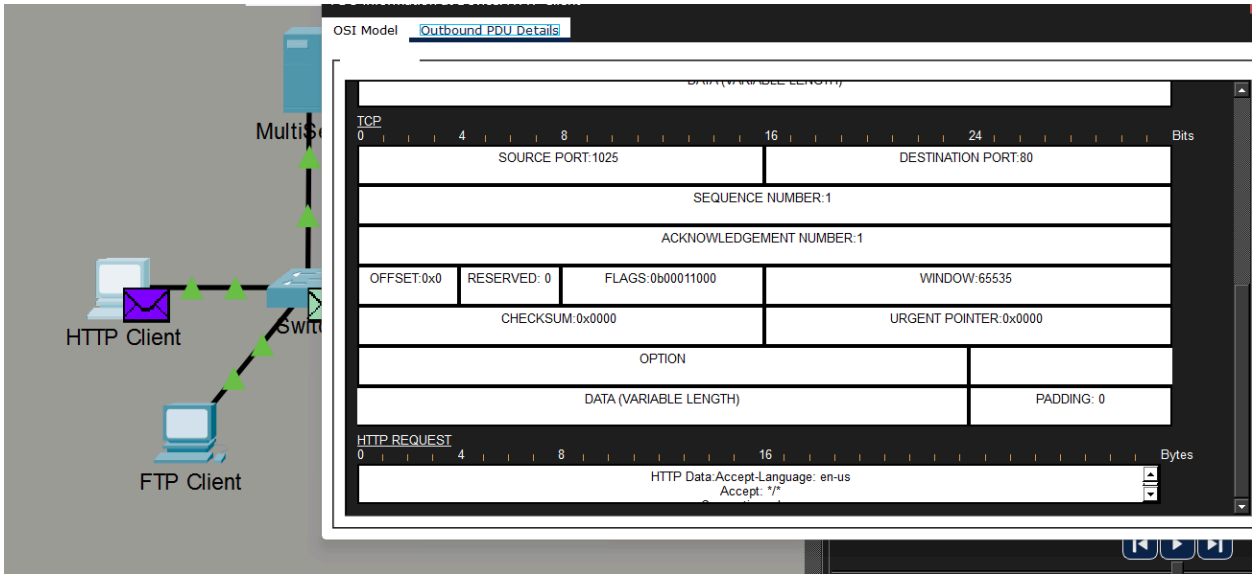
Which TCP flags are set in this PDU?
0b00011000

g. Close the PDU and click Capture/Forward until a PDU with a checkmark returns to the HTTP Client.
h. Click the PDU envelope and select Inbound PDU Details.
Question:
How are the port and sequence numbers different than before?



SOURCE PORT กับ DESTINATION PORT สลับค่ากัน เนื่องจากมีการตอบกลับจาก server สู่ client
Acknowledgement Number (ACK) เปลี่ยนค่า ACK จะระบุ หมายเลขลำดับถัดไปที่คาดว่าจะได้รับ
เป็นการยืนยันว่าได้รับข้อมูลก่อนหน้านี้เรียบร้อยแล้ว

i. Click the HTTP PDU which HTTP Client has prepared to send to MultiServer. This is the beginning of
the HTTP communication. Click this second PDU envelope and select Outbound PDU Details.
Question:
What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

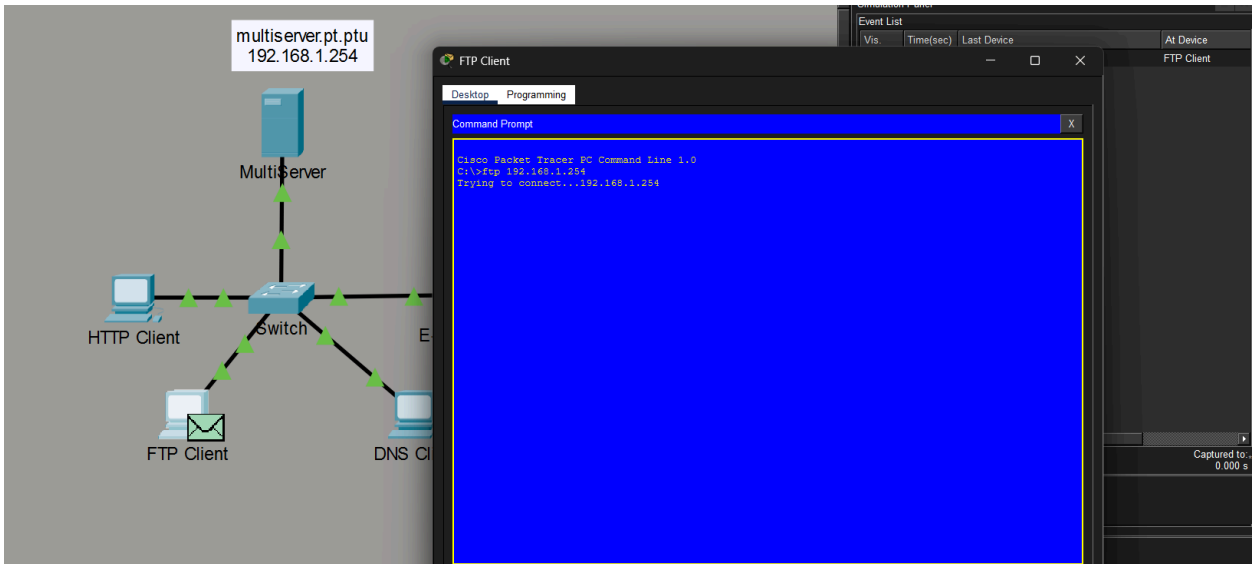


SOURCE PORT กับ DESTINATION PORT มีค่าขึ้นอยู่กับการส่งข้อมูลว่าจาก HTTP Client ไป Server หรือ Server ไป HTTP Client

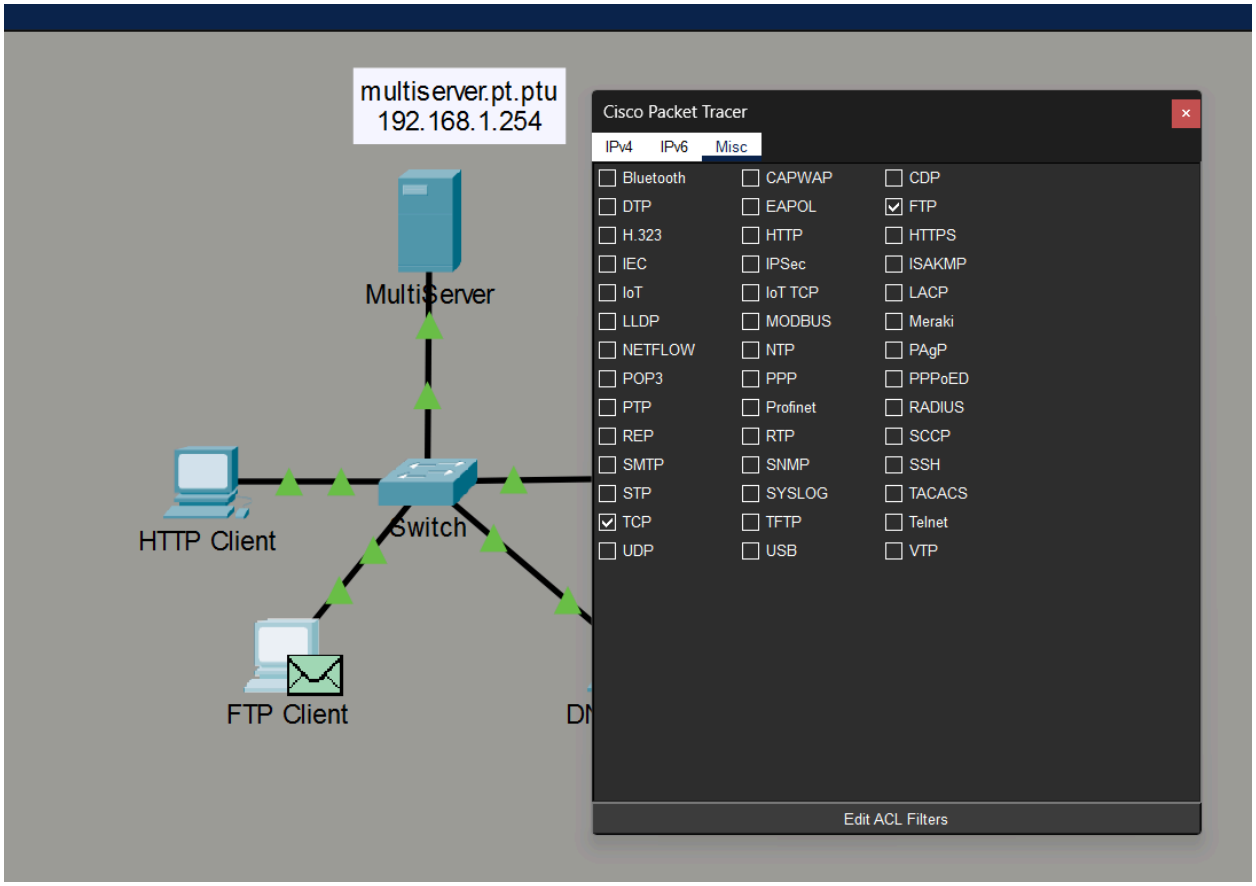
j. Reset the simulation.

Step 2: Examine FTP traffic as the clients communicate with the server.

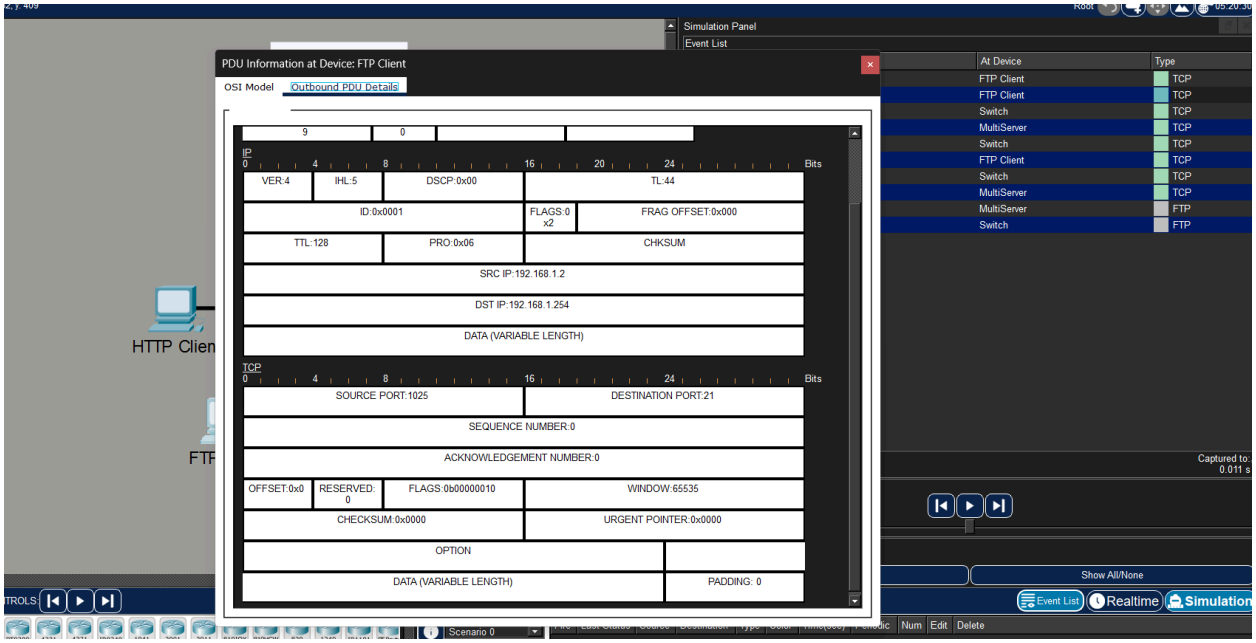
a. Open the command prompt on the FTP Client desktop. Initiate an FTP connection by entering ftp 192.168.1.254.



b. In the Simulation Panel, change Edit Filters to display only FTP and TCP.



c. Click Capture/Forward. Click the second PDU envelope to open it. Click the Outbound PDU Details tab and scroll down to the TCP section.

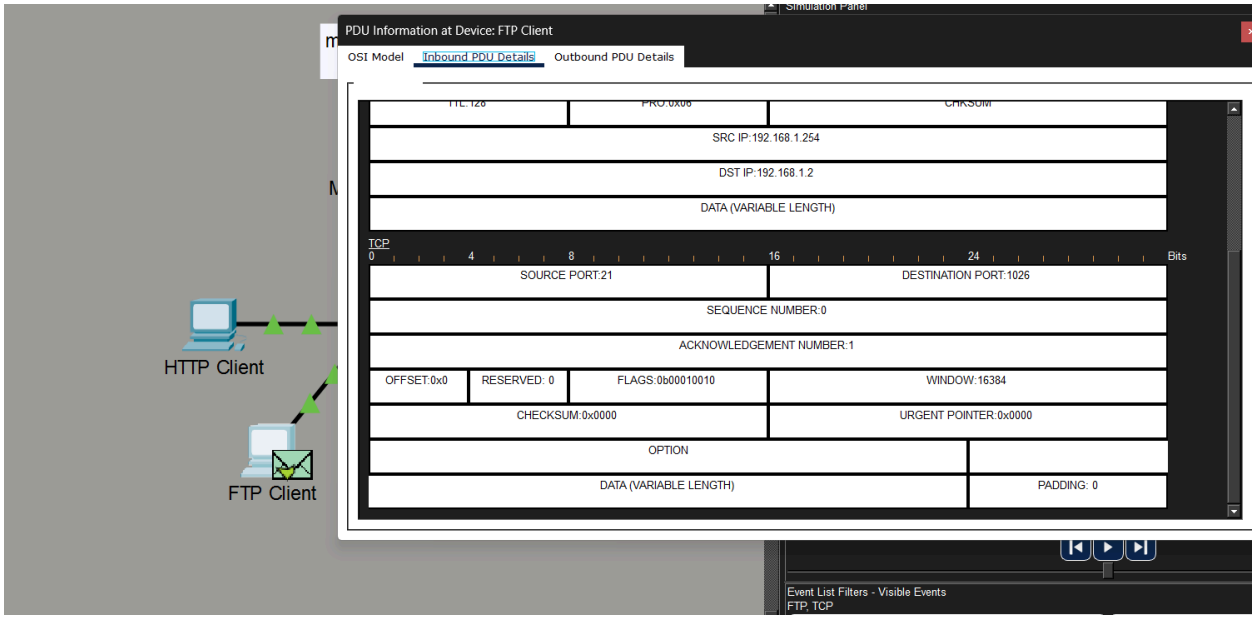


Question:

Are these communications considered to be reliable?
ใช่

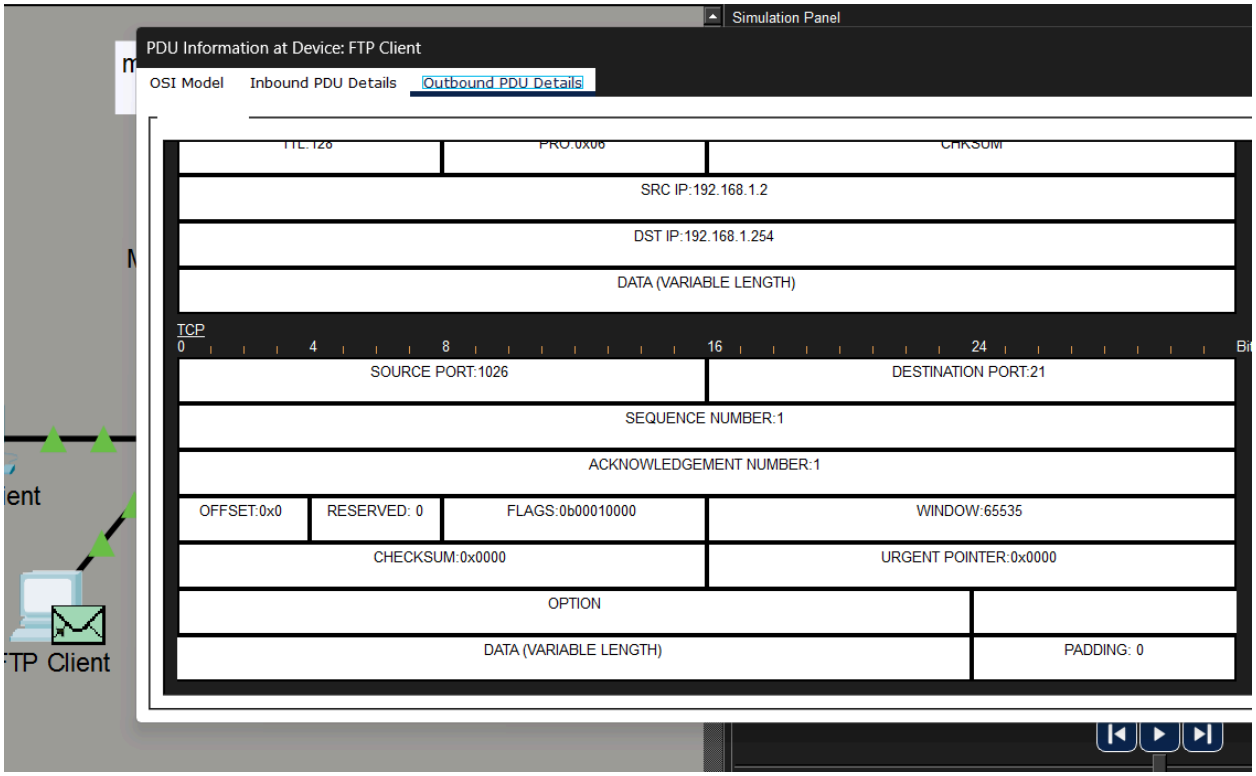
d. Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.
Question:
What is the value in the flag field?
SRC PORT : 1025
DEST PORT : 21
SEQUENCE NUM : 0
ACK NUM : 0
Flag : 0b00000010

e. Close the PDU and click Capture/Forward until a PDU returns to the FTP Client with a checkmark.
f. Click the PDU envelope and select Inbound PDU Details.
Question:
How are the port and sequence numbers different than before?



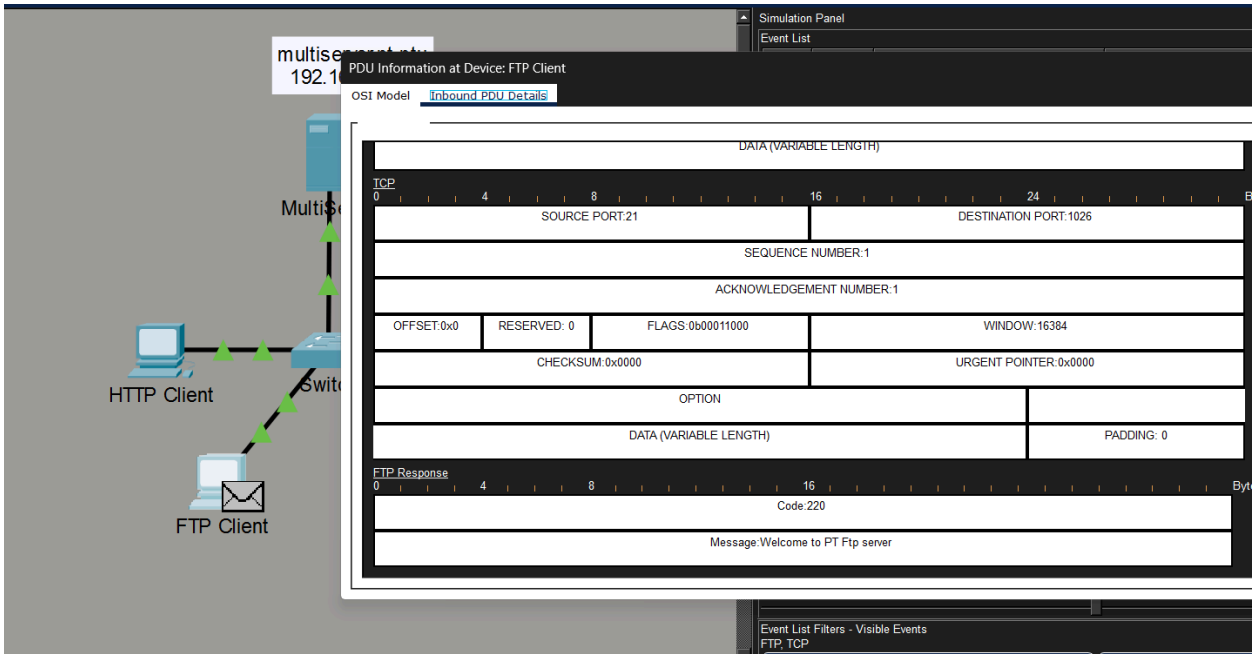
SOURCE PORT กับ DESTINATION PORT สลับค่ากัน
Sequence number : 0
ACK NUMBER : 1
FLAGS : 0b00010010

g. Click the Outbound PDU Details tab.



Question:
How are the port and sequence numbers different from the previous results?
SOURCE PORT กับ DESTINATION PORT สลับค่ากัน
SEQUENCE NUMBER : 1
FLAGS : 0b00010000

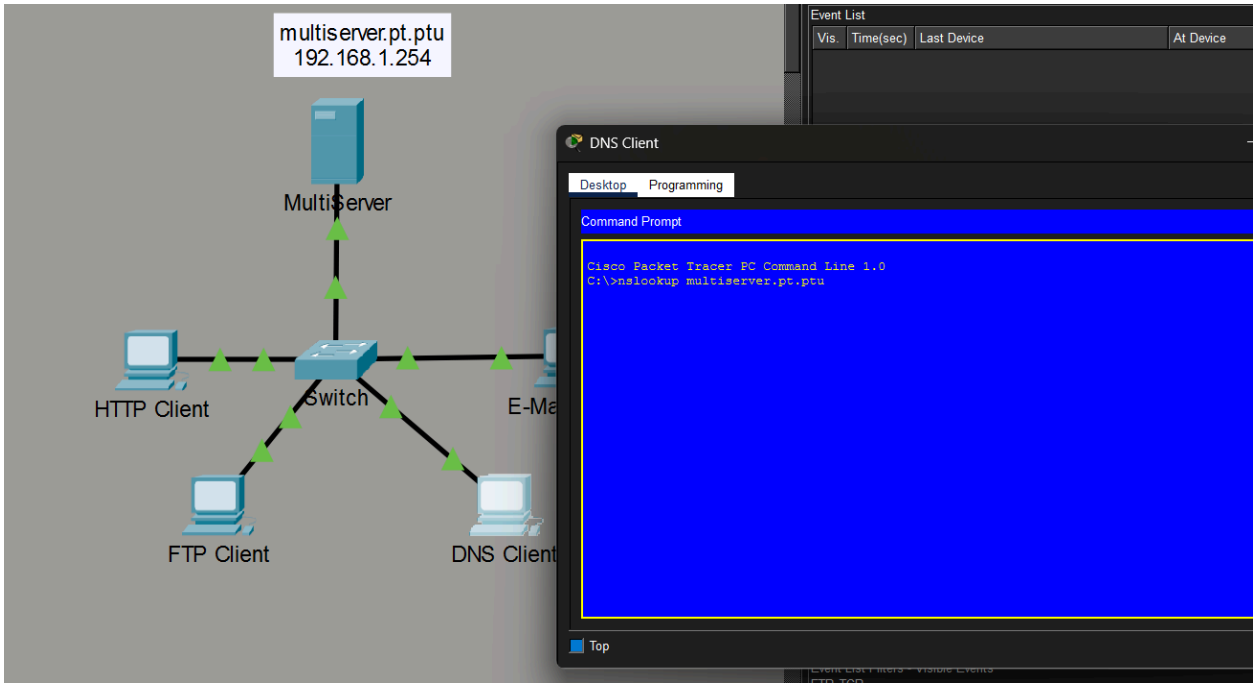
h. Close the PDU and click Capture/Forward until a second PDU returns to the FTP Client. The PDU is a
different color.
i. Open the PDU and select Inbound PDU Details. Scroll down past the TCP section.



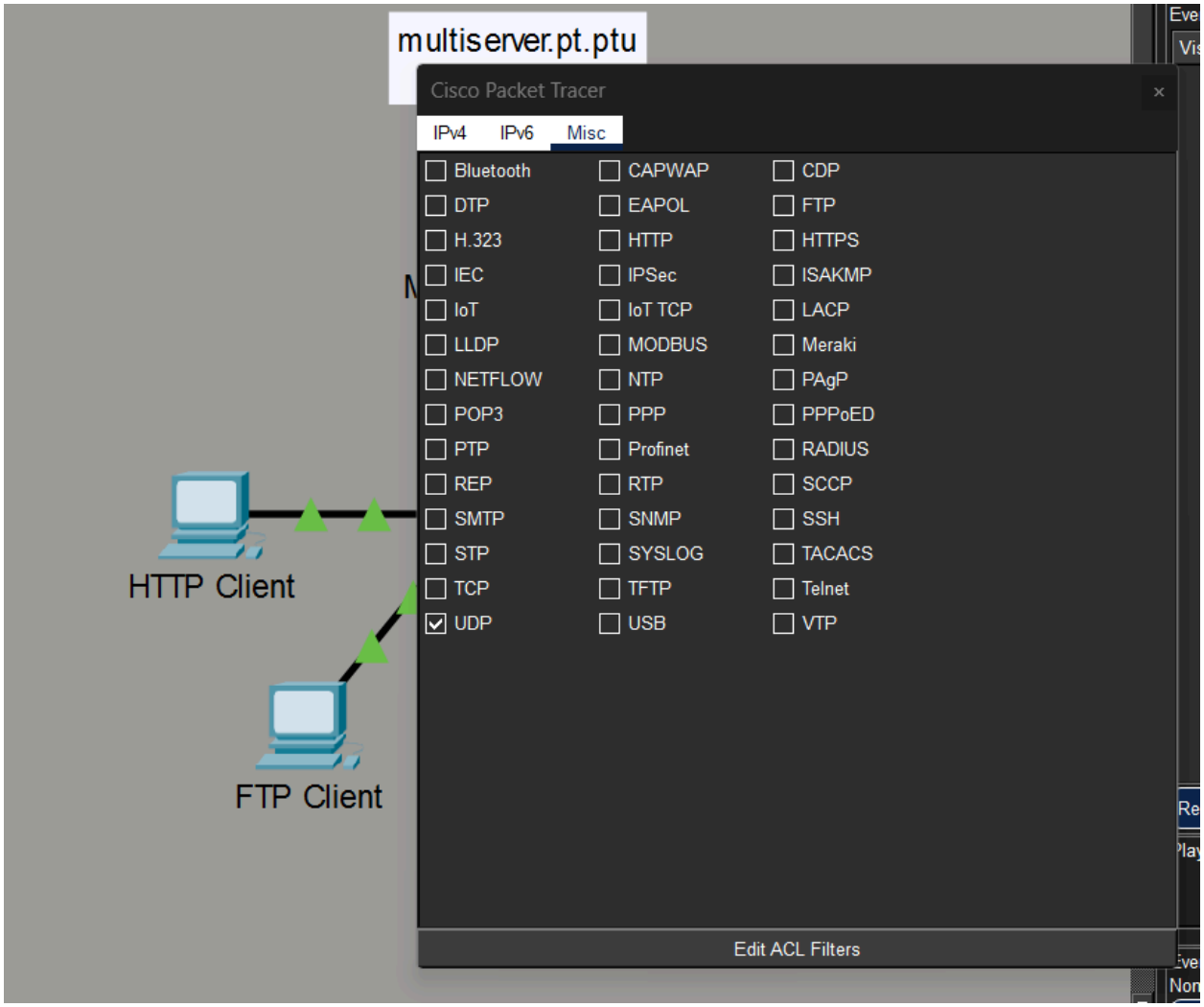
Question:
What is the message from the server?
Welcome to PT Ftp server

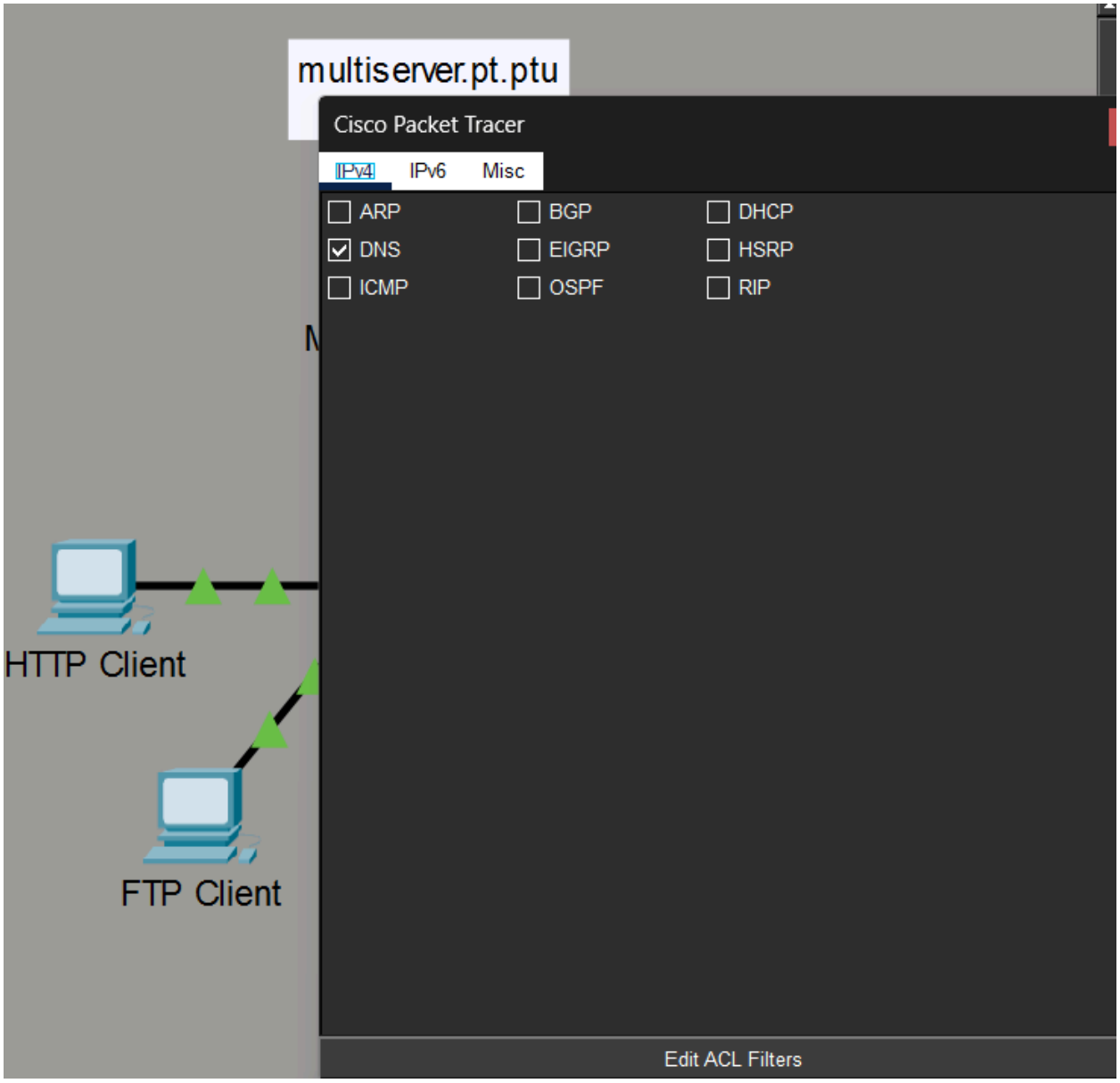
j. Click Reset Simulation.

Step 3: Examine DNS traffic as the clients communicate with the server.
a. Repeat the steps in Part 1 to create DNS traffic.

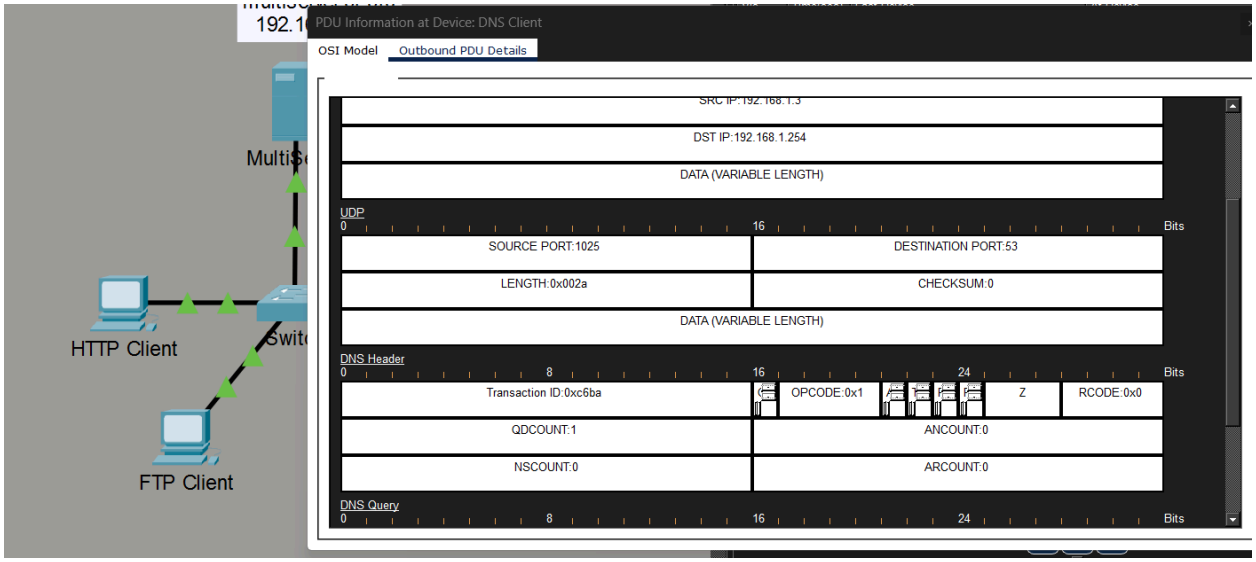


b. In the Simulation Panel, change Edit Filters to display only DNS and UDP.


multiserver.pt.ptu
Cisco Packet Tracer
IPv4
IPv6
Misc
Bluetooth
CAPWAP
CDP
DTP
EAPOL
FTP
H.323
HTTP
HTTPS
IEC
IPSec
ISAKMP
IoT
IoT TCP
LACP
LLDP
MODBUS
Meraki
NETFLOW
NTP
PAgP
POP3
PPP
PPPoED
PTP
Profinet
RADIUS
REP
RTP
SCCP
SMTP
SNMP
SSH
STP
SYSLOG
TACACS
TCP
TFTP
Telnet
UDP
USB
VTP
Edit ACL Filters
HTTP Client
FTP Client



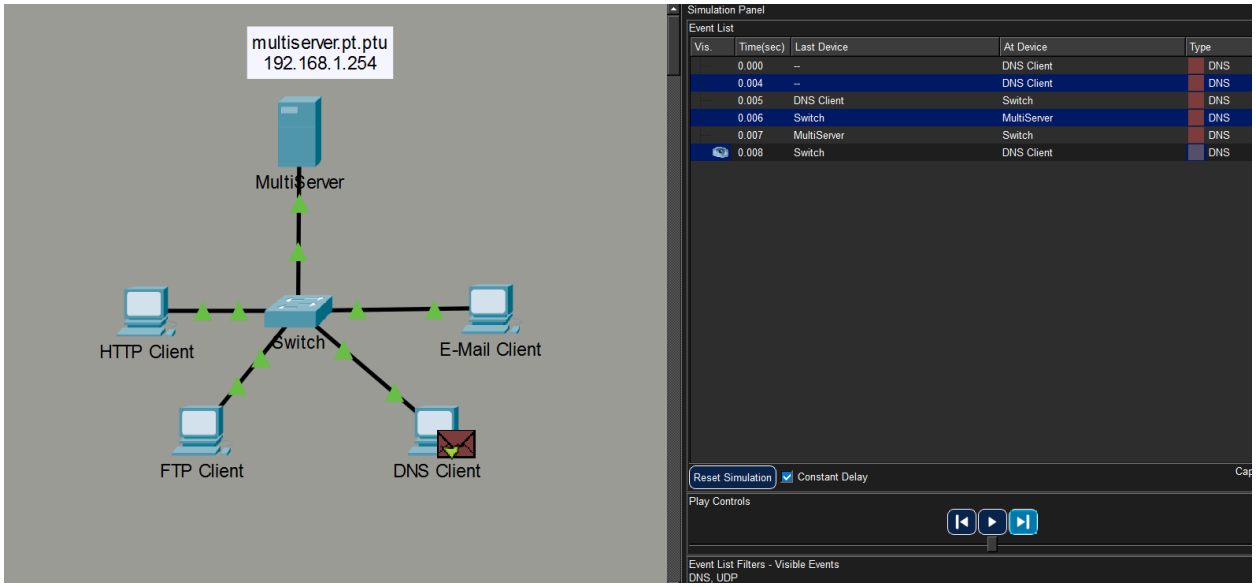
c. Click the PDU envelope to open it.



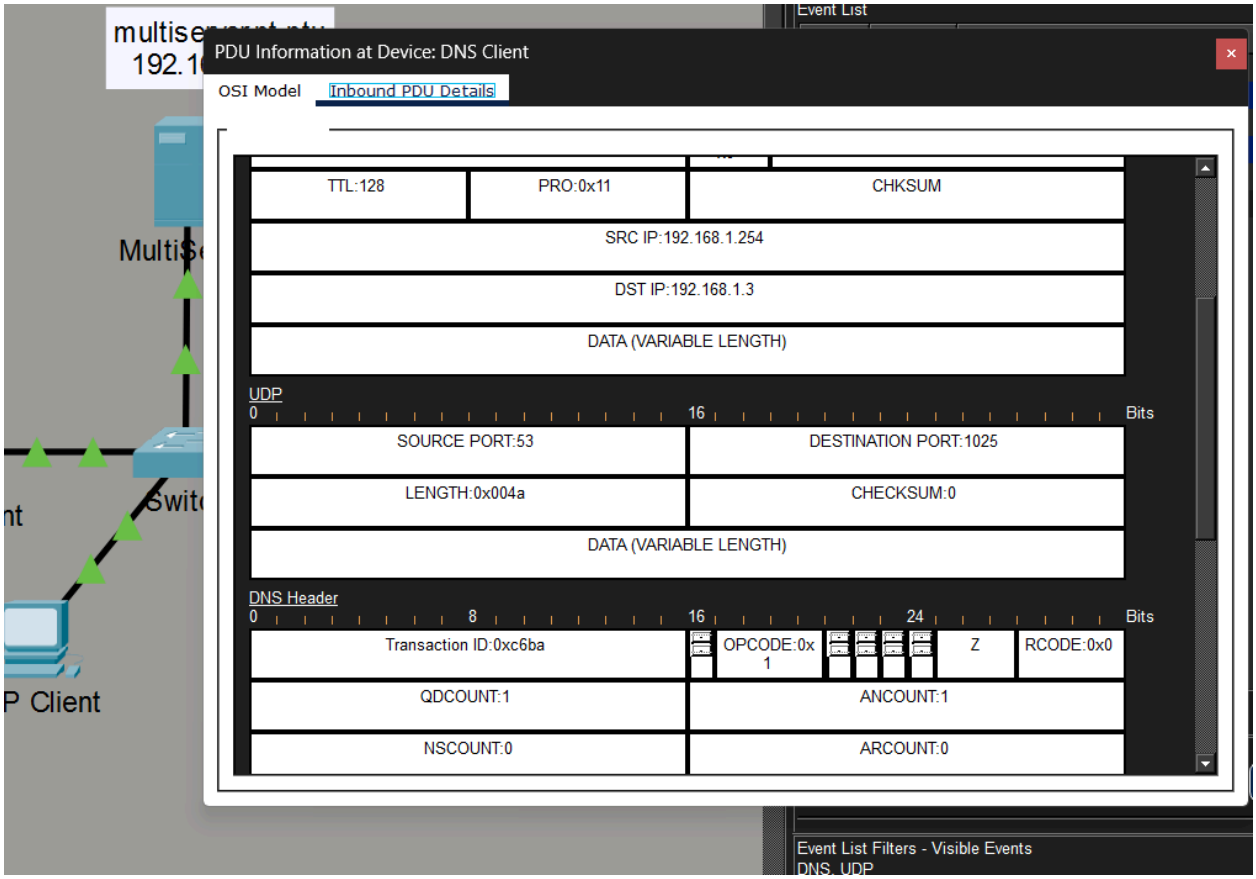
What is the Layer 4 protocol?
UDP
Are these communications considered to be reliable?
ไม่ เพราะ udp เป็น unreliable communication

e. Open the Outbound PDU Details tab and find the UDP section of the PDU formats. Record the SRC
PORT and DEST PORT values.
Question:
Why are there no sequence and acknowledgement numbers?
จะมีเมื่อใช้ reliable communicationหรือเมื่อใช้ tcp แต่ ณ ขณะนี้เป็น UDP

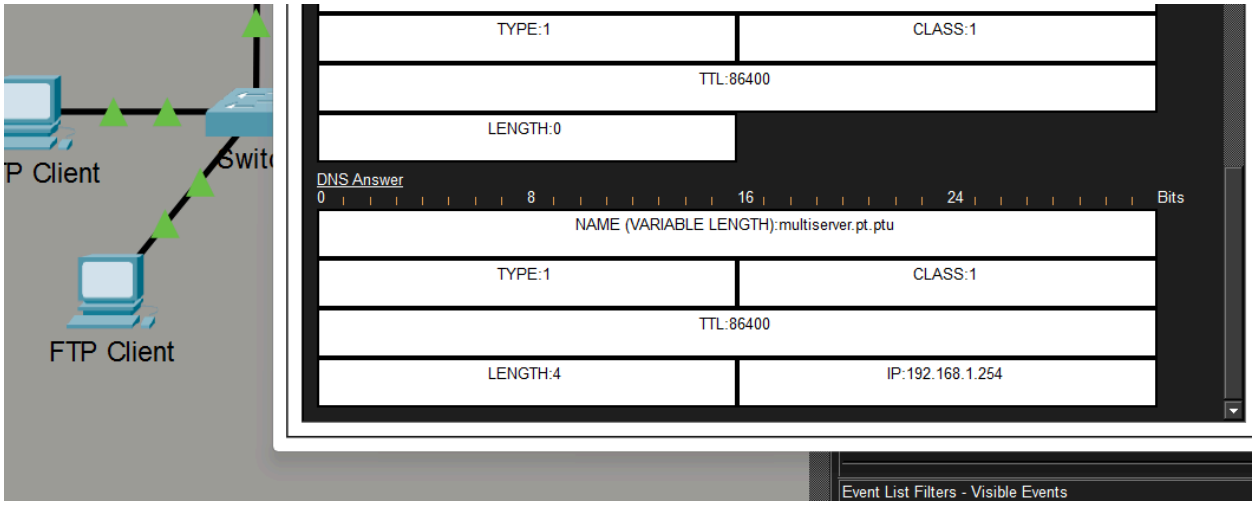
f. Close the PDU and click Capture/Forward until a PDU with a check mark returns to the DNS Client.



g. Click the PDU envelope and select Inbound PDU Details.



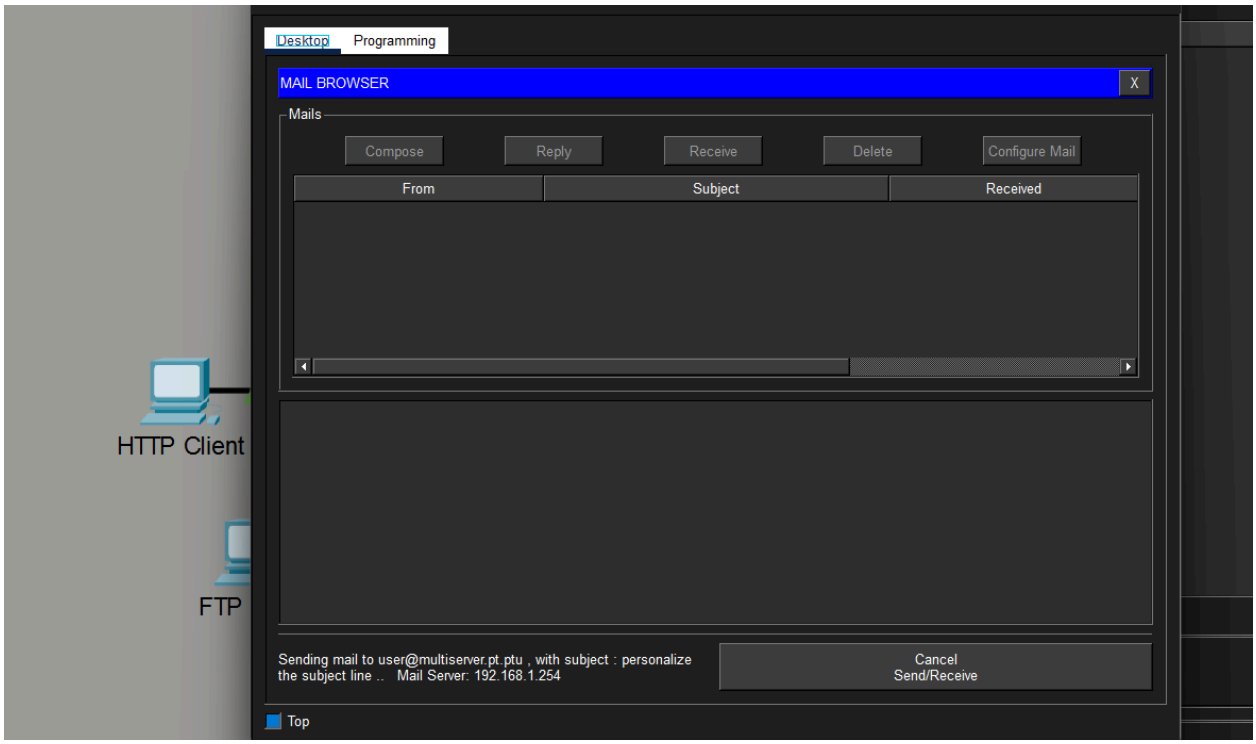
Question:
How are the port and sequence numbers different than before?
SOURCE PORT กับ DESTINATION PORT สลับค่ากัน



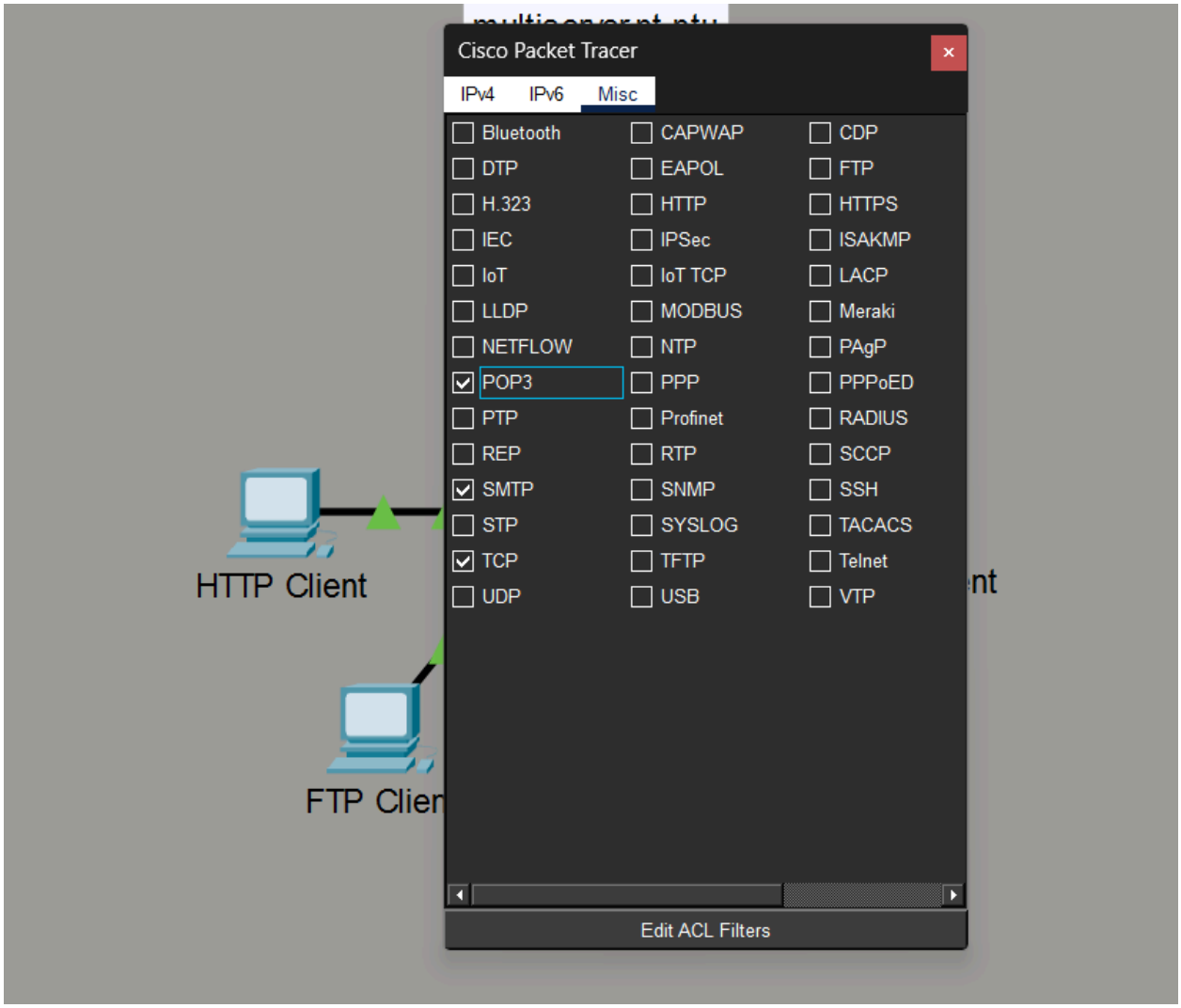
What is the last section of the PDU called? What is the IP address for the name multiserver.pt.ptu?

192.168.1.254
h. Click Reset Simulation.

Step 4: Examine email traffic as the clients communicate with the server.
a. Repeat the steps in Part 1 to send an email to user@multiserver.pt.ptu.

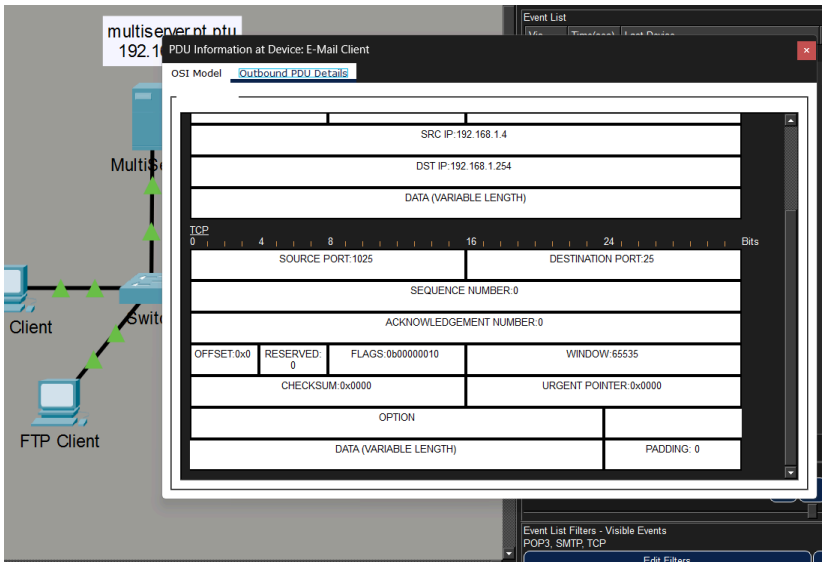


b. In the Simulation Panel, change Edit Filters to display only POP3, SMTP and TCP.



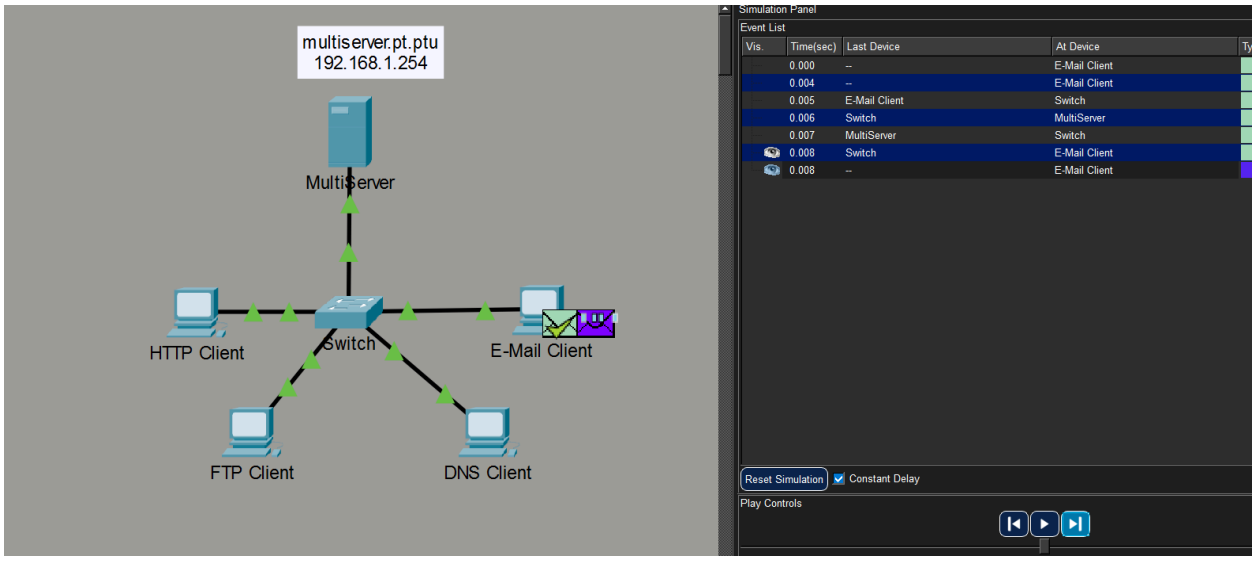
c. Click the first PDU envelope to open it.

d. Click the Outbound PDU Details tab and scroll down to the last section.

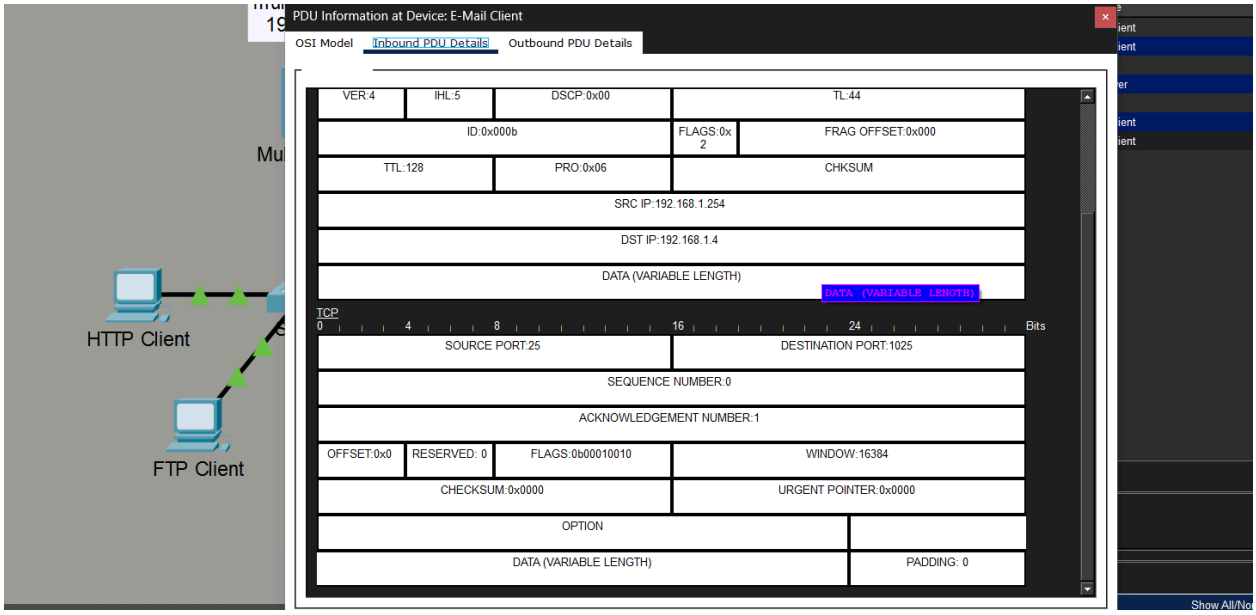


What transport layer protocol does email traffic use?
TCP
Are these communications considered to be reliable?
ใช่ เพราะ ใช้ TCP ที่เป็น reliable communication
e. Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values. What is the flag field value?
SCR PORT : 1025
DEST PORT : 25
SEQUENCE NUM : 0
ACK NUM : 0
FLAGS : 0b00000010

f. Close the PDU and click Capture/Forward until a PDU returns to the E-Mail Client with a checkmark.

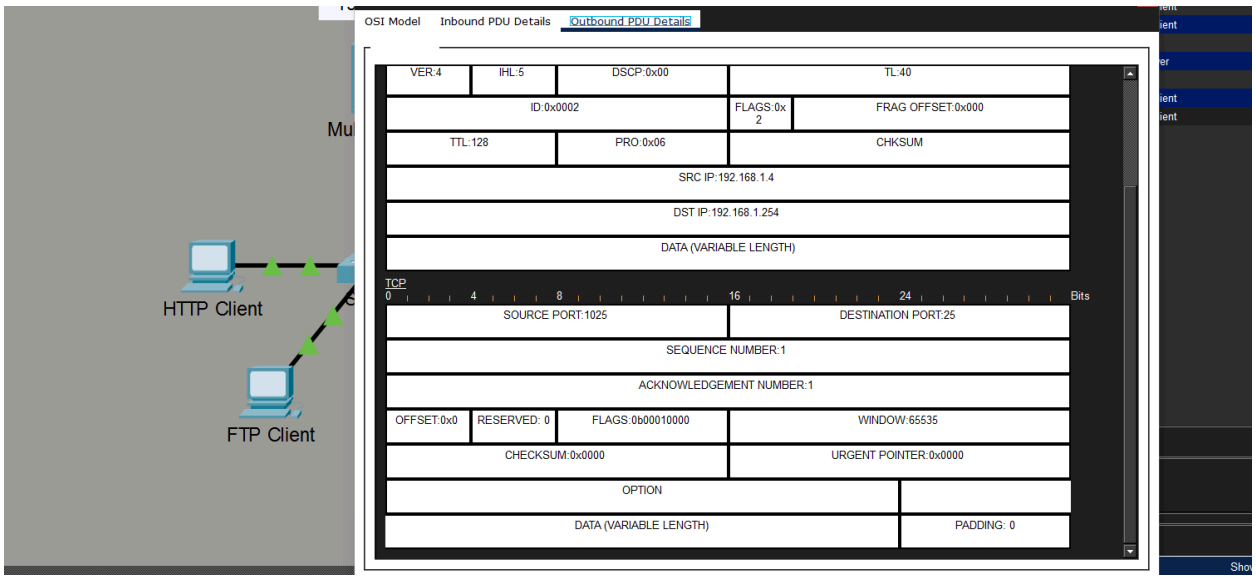


g. Click the TCP PDU envelope and select Inbound PDU Details.



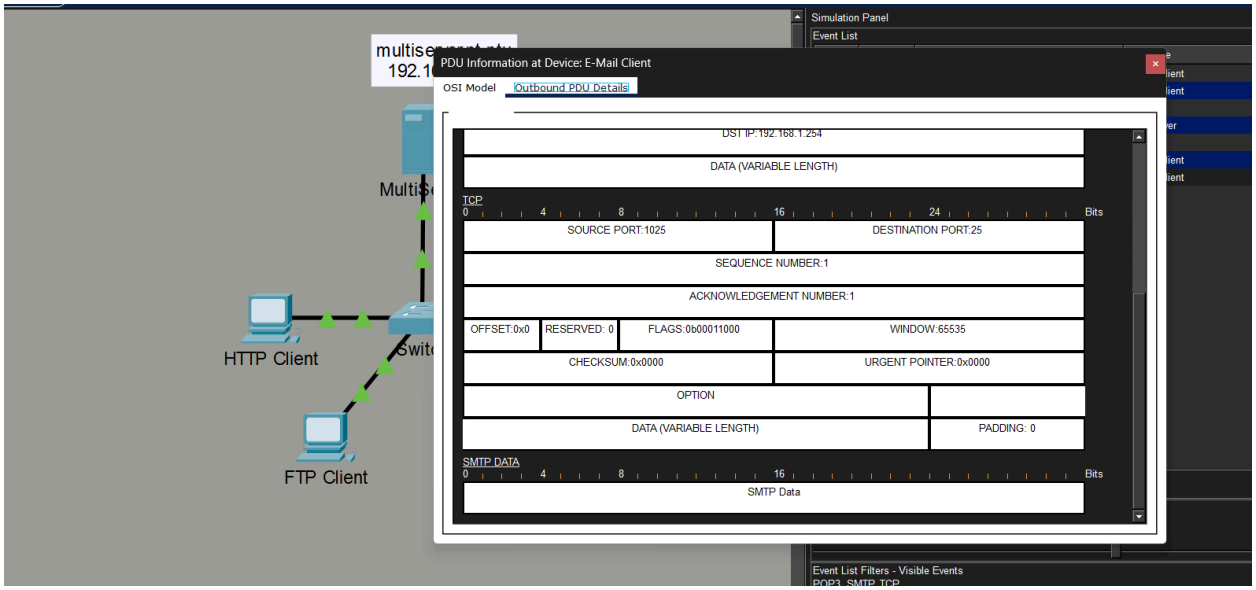
How are the port and sequence numbers different than before?
SOURCE PORT กับ DESTINATION PORT สลับค่ากัน
SEQUENCE NUMBER : 0

h. Click the Outbound PDU Details tab.



How are the port and sequence numbers different from the previous two results?
SOURCE PORT กับ DESTINATION PORT สลับค่ากัน
SEQUENCE NUMBER เพิ่มขึ้นมาเป็น 1

i. There is a second PDU of a different color that E-Mail Client has prepared to send to MultiServer. This is the beginning of the email communication. Click this second PDU envelope and select Outbound PDU Details.



How are the port and sequence numbers different from the previous two PDUs?
SOURCE PORT : 1025
DESTINATION PORT : 25
SEQUENCE NUMBER : 1

What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?
TCP port 25 ใช้กับโปรโตคอล SMTP และ TCP port 110 ใช้กับโปรโตคอล POP3